

4.7.2 Carry Generator: GATE CSE 2007 | Question: 35



In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs A_i and B_i are given by:

$$P_i = A_i \oplus B_i \text{ and } G_i = A_i B_i$$

The expressions for the sum bit S_i and the carry bit C_{i+1} of the look ahead carry adder are given by:

$$S_i = P_i \oplus C_i \text{ and } C_{i+1} = G_i + P_i C_i, \text{ where } C_0 \text{ is the input carry.}$$

Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3, S_2, S_1, S_0 and C_4 as its outputs are respectively:

- A. 6,3 B. 10,4 C. 6,4 D. 10,5

gatecse-2007 digital-logic normal carry-generator adder

Answer key

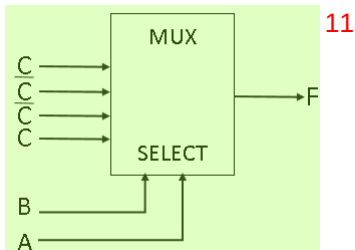
4.8 Circuit Output (40)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (8Q)

4.8.1 Circuit Output: GATE CSE 1987 | Question: 1-IV



The output F of the below multiplexer circuit can be represented by



- A. $AB + B\bar{C} + \bar{C}A + \bar{B}\bar{C}$ B. $A \oplus B \oplus C$
C. $A \oplus B$ D. $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$

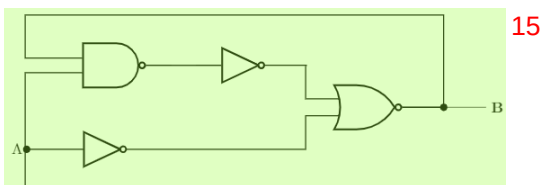
gate1987 digital-logic combinational-circuit multiplexer circuit-output

Answer key

4.8.2 Circuit Output: GATE CSE 1989 | Question: 4-ix



Explain the behaviour of the following logic circuit with level input A and output B .



gate1989 descriptive digital-logic circuit-output

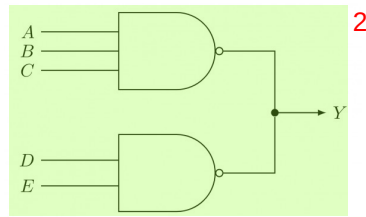
Answer key

4.8.3 Circuit Output: GATE CSE 1990 | Question: 3-i



Choose the correct alternatives (More than one may be correct).

Two NAND gates having open collector outputs are tied together as shown in below figure. 1



The logic function Y , implemented by the circuit is, 3

- A. $Y = ABC + DE$ B. $Y = \overline{ABC + DE}$ 4
C. $Y = ABC.DE$ D. $Y = \overline{ABC.DE}$

gate1990 normal digital-logic circuit-output 5

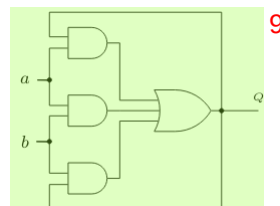
Answer key 6

4.8.4 Circuit Output: GATE CSE 1991 | Question: 5-a



Analyse the circuit in Fig below and complete the following table 7

a	b	Q_n
0	0	
0	1	
1	0	
1	1	



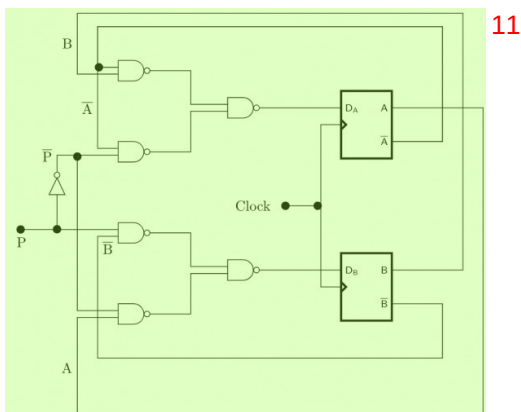
gate1991 digital-logic normal circuit-output sequential-circuit descriptive

Answer key

4.8.5 Circuit Output: GATE CSE 1993 | Question: 19



A control algorithm is implemented by the NAND – gate circuitry given in figure below, where A and B are state variable implemented by D flip-flops, and P is control input. Develop the state transition table for this controller. 10



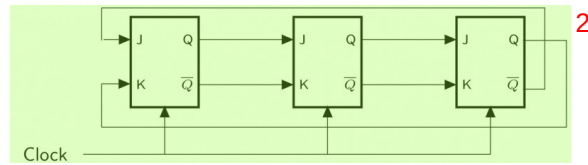
gate1993 digital-logic sequential-circuit flip-flop circuit-output normal descriptive

Answer key

4.8.6 Circuit Output: GATE CSE 1993 | Question: 6-3



For the initial state of 000, the function performed by the arrangement of the J-K flip-flops in figure is: 1



- A. Shift Register
B. Mod- 3 Counter
C. Mod- 6 Counter
D. Mod- 2 Counter
E. None of the above

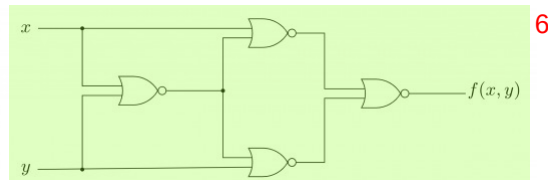
gate1993 digital-logic sequential-circuit flip-flop digital-counter circuit-output multiple-selects 4

Answer key

4.8.7 Circuit Output: GATE CSE 1993 | Question: 6.1



Identify the logic function performed by the circuit shown in figure. 5



- A. exclusive OR
B. exclusive NOR
C. NAND
D. NOR
E. None of the above

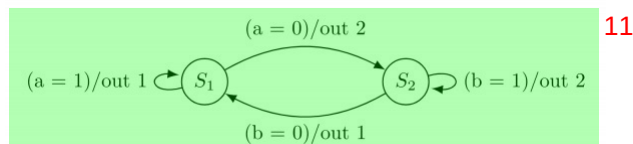
gate1993 digital-logic combinational-circuit circuit-output normal 8

Answer key 9

4.8.8 Circuit Output: GATE CSE 1993 | Question: 6.2



If the state machine described in figure should have a stable state, the restriction on the inputs is given by 10



- A. $a \cdot b = 1$
B. $a + b = 1$
C. $\bar{a} + \bar{b} = 0$
D. $\overline{a \cdot b} = 1$
E. $\overline{a + b} = 1$

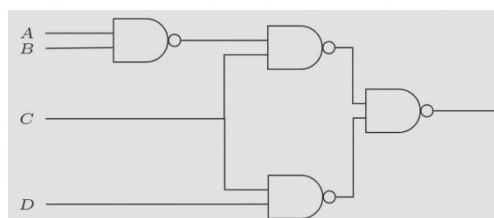
gate1993 digital-logic normal circuit-output sequential-circuit 13

Answer key 14

4.8.9 Circuit Output: GATE CSE 1994 | Question: 1.8



The logic expression for the output of the circuit shown in figure below is:



A. $\overline{AC} + \overline{BC} + CD$

B. $\overline{AC} + \overline{BC} + CD$ ¹

C. $ABC + \overline{C}\overline{D}$

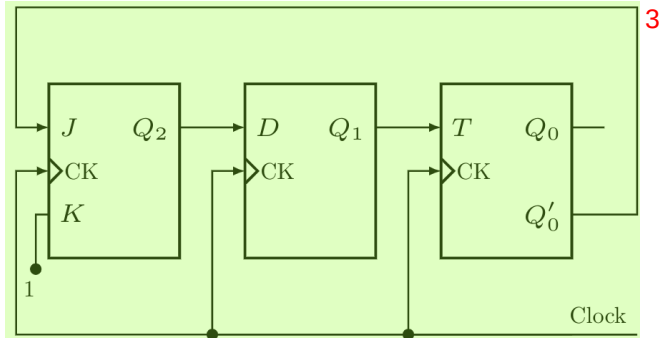
D. $\overline{A}\overline{B} + \overline{B}\overline{C} + CD$

gate1994 digital-logic circuit-output normal

Answer key

4.8.10 Circuit Output: GATE CSE 1994 | Question: 11

Find the contents of the flip-flop Q_2, Q_1 and Q_0 in the circuit of figure, after giving four clock pulses to the clock terminal. Assume $Q_2Q_1Q_0 = 000$ initially. ²

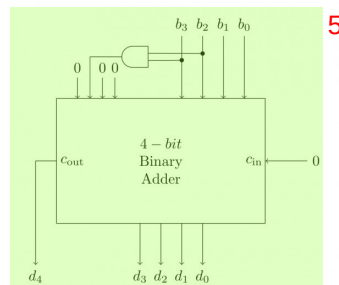


gate1994 digital-logic sequential-circuit digital-counter circuit-output normal descriptive

Answer key

4.8.11 Circuit Output: GATE CSE 1996 | Question: 2.21

Consider the circuit in below figure which has a four bit binary number $b_3b_2b_1b_0$ as input and a five bit binary number, $d_4d_3d_2d_1d_0$ as output. ⁴



A. Binary to Hex conversion

B. Binary to BCD conversion

C. Binary to Gray code conversion

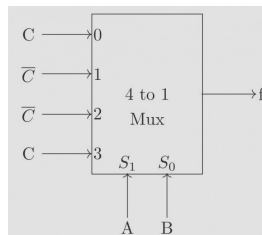
D. Binary to *radix* - 12 conversion ⁶

gate1996 digital-logic circuit-output normal ⁷

Answer key

4.8.12 Circuit Output: GATE CSE 1996 | Question: 2.22

Consider the circuit in figure. f implements

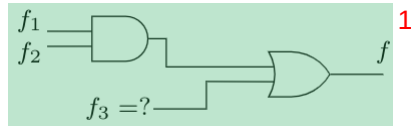


A. $\overline{ABC} + \overline{ABC} + ABC$

B. $A + B + C$

C. $A \oplus B \oplus C$

D. $AB + BC + CA$



The function f_3 is ²

- A. $\sum 9, 10$ B. $\sum 9$ C. $\sum 1, 8, 9$ D. $\sum 8, 10, 15$ ³

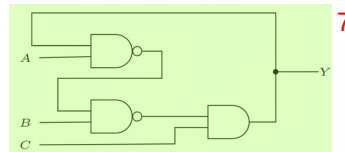
gate1997 digital-logic circuit-output normal ⁴

Answer key ⁵

4.8.16 Circuit Output: GATE CSE 1999 | Question: 2.8



Consider the circuit shown below. In a certain steady state, the line Y is at '1'. What are the possible values of A, B and C in this state? ⁶



- A. $A = 0, B = 0, C = 1$ B. $A = 0, B = 1, C = 1$ ⁸
C. $A = 1, B = 0, C = 1$ D. $A = 1, B = 1, C = 1$

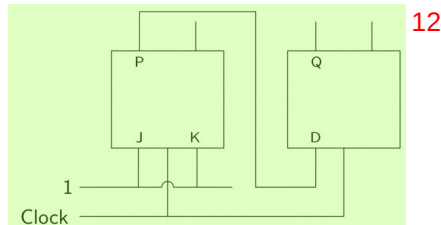
gate1999 digital-logic circuit-output normal ⁹

Answer key ¹⁰

4.8.17 Circuit Output: GATE CSE 2000 | Question: 2.12



The following arrangement of master-slave flip flops ¹¹



has the initial state of P, Q as 0, 1 (respectively). After three clock cycles the output state P, Q is (respectively), ¹³

- A. 1, 0 B. 1, 1 C. 0, 0 D. 0, 1 ¹⁴

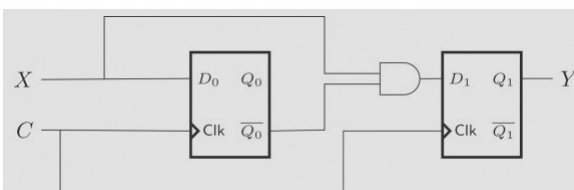
gatecse-2000 digital-logic circuit-output normal flip-flop ¹⁵

Answer key ¹⁶

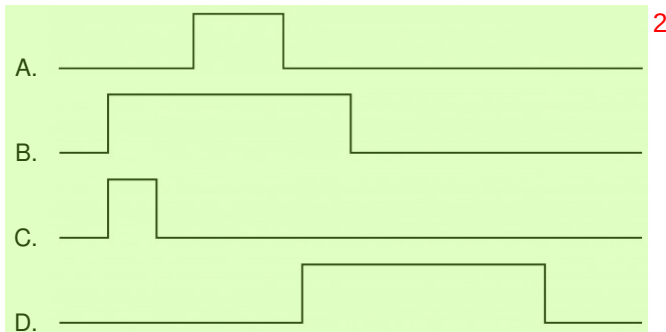
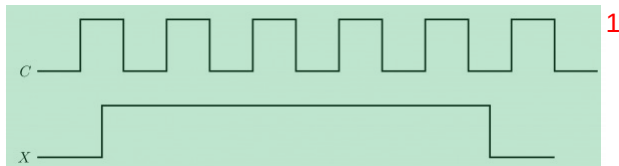
4.8.18 Circuit Output: GATE CSE 2001 | Question: 2.8



Consider the following circuit with initial state $Q_0 = Q_1 = 0$. The D Flip-flops are positive edged triggered and have set up times 20 nanosecond and hold times 0.



Consider the following timing diagrams of X and C . The clock period of $C \geq 40$ nanosecond. Which one is the correct plot of Y ?

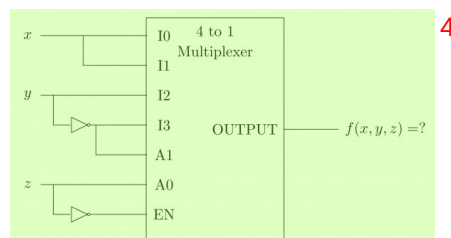


gatecse-2001 digital-logic circuit-output normal

Answer key

4.8.19 Circuit Output: GATE CSE 2002 | Question: 2.2

Consider the following multiplexer where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$ respectively and f is the output of the multiplexor. EN is the Enable input.



The function $f(x, y, z)$ implemented by the above circuit is

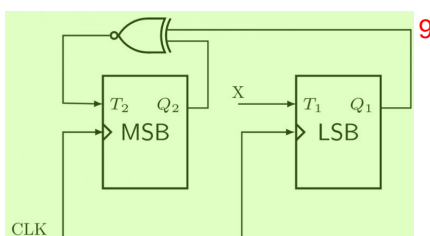
- A. xyz' B. $xy + z$ C. $x + y$ D. None of the above

gatecse-2002 digital-logic circuit-output normal

Answer key

4.8.20 Circuit Output: GATE CSE 2004 | Question: 61

Consider the partial implementation of a 2-bit counter using T flip-flops following the sequence 0-2-3-1-0, as shown below.



To complete the circuit, the input X should be

- A. Q_2^c B. $Q_2 + Q_1$ C. $(Q_1 + Q_2)^c$ D. $Q_1 \oplus Q_2$

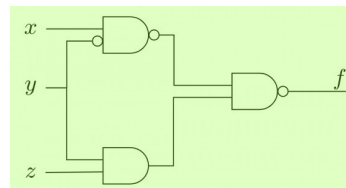
gatecse-2004 digital-logic circuit-output normal

Answer key

4.8.21 Circuit Output: GATE CSE 2005 | Question: 15



Consider the following circuit: 1



Which one of the following is TRUE? 3

- | | |
|------------------------------|-----------------------------------|
| A. f is independent of x | B. f is independent of y |
| C. f is independent of z | D. None of x, y, z is redundant |

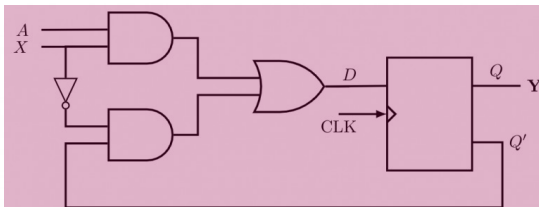
gatesee-2005 digital-logic circuit-output normal 5

Answer key

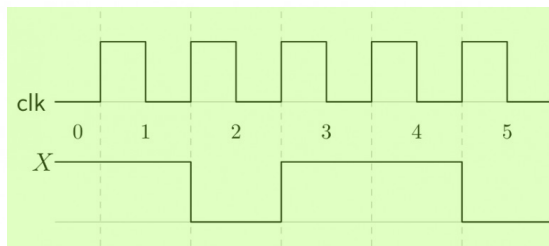
4.8.22 Circuit Output: GATE CSE 2005 | Question: 62



Consider the following circuit involving a positive edge triggered D FF. 6



Consider the following timing diagram. Let A_i represents the logic level on the line A in the i -th clock period. 8



Let A' represent the complement of A . The correct output sequence on Y over the clock periods 1 through 5 is: 10

- | | |
|---------------------------|----------------------------|
| A. $A_0 A_1 A'_1 A_3 A_4$ | B. $A_0 A_1 A'_2 A_3 A_4$ |
| C. $A_1 A_2 A'_2 A_3 A_4$ | D. $A_1 A'_2 A_3 A_4 A'_5$ |

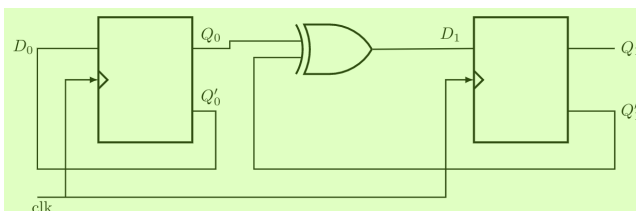
gatesee-2005 digital-logic circuit-output normal 12

Answer key

4.8.23 Circuit Output: GATE CSE 2005 | Question: 64



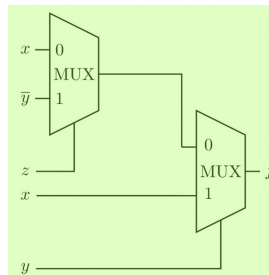
Consider the following circuit: 13



The flip-flops are positive edge triggered D FFs. Each state is designated as a two-bit string $Q_0 Q_1$. Let the initial state be 00. The state transition sequence is 15

- | | | | |
|---------------------------------------|------------------------|--|--|
| A. $00 \rightarrow 11 \rightarrow 01$ | B. $00 \rightarrow 11$ | C. $00 \rightarrow 10 \rightarrow 01 \rightarrow 11$ | D. $00 \rightarrow 11 \rightarrow 01 \rightarrow 10$ |
|---------------------------------------|------------------------|--|--|

4.8.24 Circuit Output: GATE CSE 2006 | Question: 35



1

Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$ 2

A. $x\bar{z} + xy + \bar{y}z$

B. $x\bar{z} + xy + \bar{y}\bar{z}$ 3

C. $xz + xy + \bar{y}\bar{z}$

D. $xz + x\bar{y} + \bar{y}z$

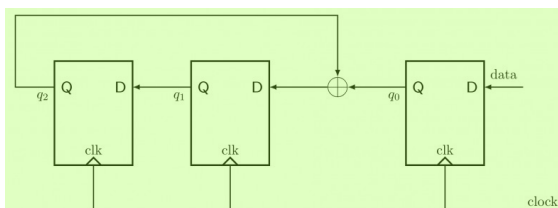
gatecse-2006 digital-logic circuit-output normal 4

Answer key 5

4.8.25 Circuit Output: GATE CSE 2006 | Question: 37



Consider the circuit in the diagram. The $\oplus$ operator represents Ex-OR. The D flip-flops are initialized to zeroes (cleared).



7

The following data: 100110000 is supplied to the "data" terminal in nine clock cycles. After that the values of $q_2q_1q_0$ are: 8

A. 000

B. 001

C. 010

D. 101 9

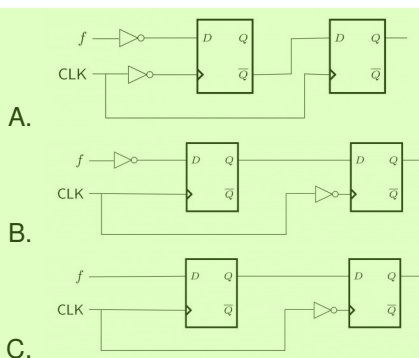
gatecse-2006 digital-logic circuit-output easy 10

Answer key

4.8.26 Circuit Output: GATE CSE 2006 | Question: 8



You are given a free running clock with a duty cycle of 50% and a digital waveform f which changes only at the negative edge of the clock. Which one of the following circuits (using clocked D flip-flops) will delay the phase of f by 180° ?



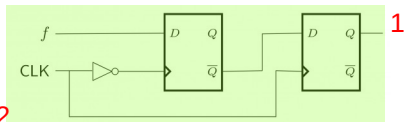
12

A.

B.

C.

11



D.2

gatecse-2006 digital-logic normal circuit-output

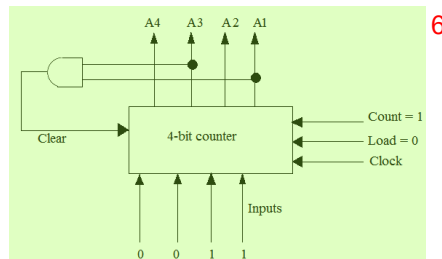
Answer key

4.8.27 Circuit Output: GATE CSE 2007 | Question: 36

The control signal functions of a 4-bit binary counter are given below (where X is "don't care"): 3

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No Change
0	↑	1	X	Load Input
0	↑	0	1	Count Next

The counter is connected as follows: 5



Assume that the counter and gate delays are negligible. If the counter starts at 0, then it cycles through the following sequence: 7

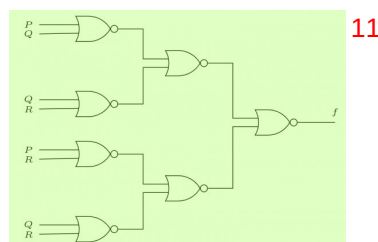
- A. 0,3,4 B. 0,3,4,5 C. 0,1,2,3,4 D. 0,1,2,3,4,5 8

gatecse-2007 digital-logic circuit-output normal 9

Answer key

4.8.28 Circuit Output: GATE CSE 2010 | Question: 31

What is the boolean expression for the output f of the combinational logic circuit of NOR gates given below? 15



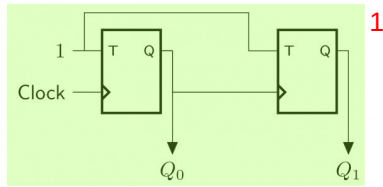
- A. $\overline{Q + R}$ B. $\overline{P + Q}$ C. $\overline{P + R}$ D. $\overline{P + Q + R}$ 12

gatecse-2010 digital-logic circuit-output normal

Answer key 13

4.8.29 Circuit Output: GATE CSE 2010 | Question: 32

In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00. What are the next four values of Q_1Q_0 ? 16



- A. 11, 10, 01, 00
C. 10, 00, 01, 11

- B. 10, 11, 01, 00
D. 11, 10, 00, 01

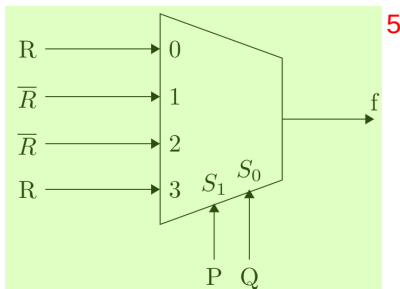
gatecse-2010 digital-logic circuit-output normal 3

Answer key

4.8.30 Circuit Output: GATE CSE 2010 | Question: 9



The Boolean expression of the output f of the multiplexer shown below is 4



- A. $\overline{P \oplus Q \oplus R}$
C. $P + Q + R$

- B. $P \oplus Q \oplus R$
D. $\overline{P + Q + R}$

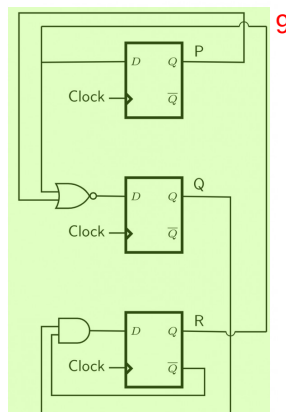
gatecse-2010 digital-logic circuit-output easy multiplexer 7

Answer key

4.8.31 Circuit Output: GATE CSE 2011 | Question: 50



Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration. 8



If at some instance prior to the occurrence of the clock edge, P, Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge? 10

A. 000

B. 001

C. 010

D. 011 11

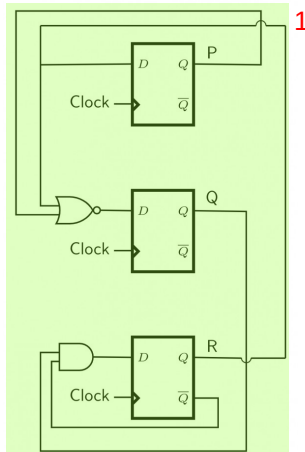
gatecse-2011 digital-logic circuit-output flip-flop normal 12

Answer key

4.8.32 Circuit Output: GATE CSE 2011 | Question: 51



Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.



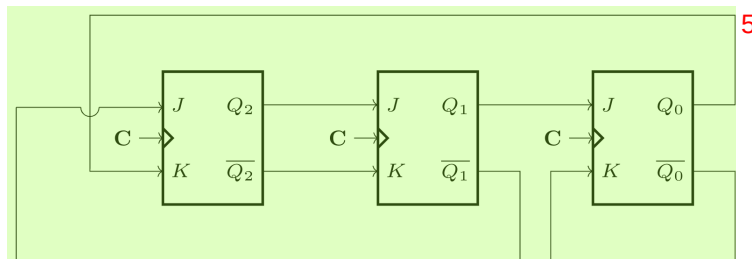
If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter? 2

- A. 3 B. 4 C. 5 D. 6 3

gatecse-2011 digital-logic circuit-output normal 4

Answer key

4.8.33 Circuit Output: GATE CSE 2014 | Set 3 | Question: 45



The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is 6

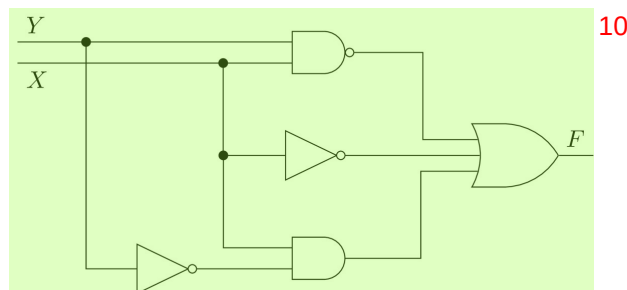
- A. 001,010,011 B. 111,110,101 7
C. 100,110,111 D. 100,011,001

gatecse-2014-set3 digital-logic circuit-output normal 8

Answer key

4.8.34 Circuit Output: GATE CSE 2025 | Set 2 | Question: 21

Consider the following logic circuit diagram. 9



Which is/are the CORRECT option(s) for the output function F ? 11

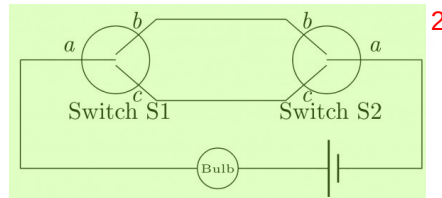
- A. $\overline{XY}$ B. $\overline{X} + \overline{Y} + XY$ 12
C. $\overline{XY} + \overline{X} + XY$ D. $X + \overline{Y}$

gatecse2025-set2 digital-logic circuit-output multiple-selects easy one-mark 13

4.8.35 Circuit Output: GATE IT 2005 | Question: 10



A two-way switch has three terminals a , b and c . In ON position (logic value 1), a is connected to b , and in OFF position, a is connected to c . Two of these two-way switches $S1$ and $S2$ are connected to a bulb as shown below.



Which of the following expressions, if true, will always result in the lighting of the bulb ?

- A. $S1 \cdot \overline{S2}$ B. $S1 + S2$
C. $S1 \oplus S2$ D. $S1 \oplus S2$

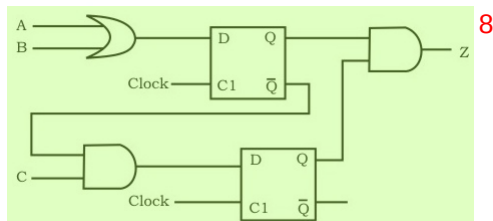
gateit-2005 digital-logic circuit-output normal 5

Answer key

4.8.36 Circuit Output: GATE IT 2005 | Question: 43



Which of the following input sequences will always generate a 1 at the output z at the end of the third cycle?



- A.

A	B	C
0	0	0
1	0	1
1	1	1

 B.

A	B	C
1	0	1
1	1	0
1	1	1

 C.

A	B	C
0	1	1
1	0	1
1	1	1

 D.

A	B	C
0	0	1
1	1	0
1	1	1

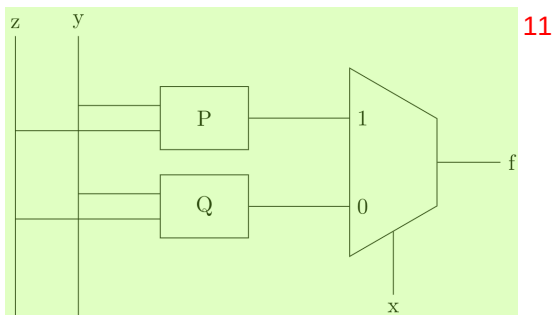
gateit-2005 digital-logic circuit-output normal 9

Answer key

4.8.37 Circuit Output: GATE IT 2006 | Question: 36



The majority function is a Boolean function $f(x, y, z)$ that takes the value 1 whenever a majority of the variables x, y, z are 1. In the circuit diagram for the majority function shown below, the logic gates for the boxes labeled P and Q are, respectively,



- A. XOR, AND B. XOR, XOR C. OR, OR D. OR, AND

12

Answer key

4.8.38 Circuit Output: GATE IT 2007 | Question: 38

The following expression was to be realized using 2-input AND and OR gates. However, during the fabrication all 2-input AND gates were mistakenly substituted by 2-input NAND gates.

$$(a.b).c + (a'.c).d + (b.c).d + a.d$$

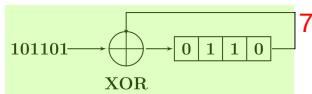
What is the function finally realized ? 2

- A. 1 B. $a' + b' + c' + d'$ 3
C. $a' + b + c' + d'$ D. $a' + b' + c + d'$

Answer key

4.8.39 Circuit Output: GATE IT 2007 | Question: 40

What is the final value stored in the linear feedback shift register if the input is 101101? 6

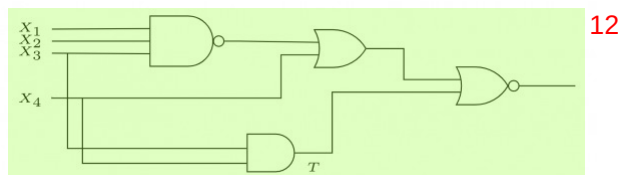


- A. 0110 B. 1011 C. 1101 D. 1111 8

Answer key 10

4.8.40 Circuit Output: GATE IT 2007 | Question: 45

The line T in the following figure is permanently connected to the ground. 11



Which of the following inputs ($X_1X_2X_3X_4$) will detect the fault ? 13

- A. 0000 B. 0111 C. 1111 D. None of these 14

Answer key 16

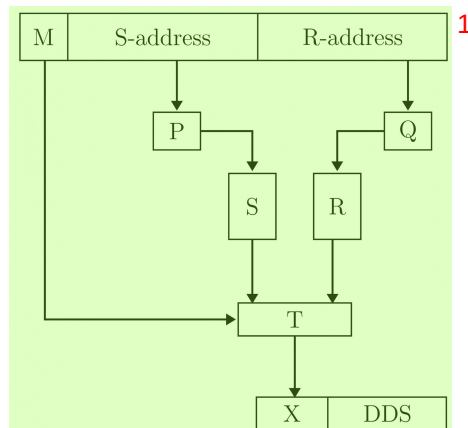
4.9

Combinational Circuit (2)

Practice Test: Test 1 (7Q)

4.9.1 Combinational Circuit: GATE CSE 2022 | Question: 30

Consider a digital display system (DDS) shown in the figure that displays the contents of register X . A 16-bit code word is used to load a word in X , either from S or from R . S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M , T selects an input word to load in X . P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P , Q , and T ?



1

- A. P is 10 : 1 multiplexer; Q is 5 : 1 multiplexer; T is 2 : 1 multiplexer
 B. P is 10 : 2^{10} decoder; Q is 5 : 2^5 decoder; T is 2 : 1 encoder
 C. P is 10 : 2^{10} decoder; Q is 5 : 2^5 decoder; T is 2 : 1 multiplexer
 D. P is 1 : 10 de-multiplexer; Q is 1 : 5 de-multiplexer; T is 2 : 1 multiplexer

2

gatecse-2022 digital-logic combinational-circuit two-marks

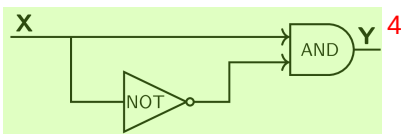
Answer key

4.9.2 Combinational Circuit: GATE CSE 2024 | Set 1 | Question: 18

Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



3



4

- A. With no propagation delays, the output Y is always logic Zero
 B. With no propagation delays, the output Y is always logic One
 C. With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One
 D. With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

5

gatecse-2024-set1 multiple-selects digital-logic combinational-circuit one-mark

Answer key

4.10

Conjunctive Normal Form (1)

4.10.1 Conjunctive Normal Form: GATE CSE 2007 | Question: 48

Which of the following is TRUE about formulae in Conjunctive Normal Form?



- A. For any formula, there is a truth assignment for which at least half the clauses evaluate to true.
 B. For any formula, there is a truth assignment for which all the clauses evaluate to true.
 C. There is a formula such that for each truth assignment, at most one-fourth of the clauses evaluate to true.
 D. None of the above.

gatecse-2007 digital-logic normal conjunctive-normal-form canonical-normal-form

Answer key

4.11

Decoder (3)

Practice Test: Test 1 (7Q)

4.11.1 Decoder: GATE CSE 2007 | Question: 8, ISRO2011-31



How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

- A. 7 B. 8 C. 9 D. 10

gatecse-2007 digital-logic normal isro2011 decoder 3

Answer key

4.11.2 Decoder: GATE CSE 2020 | Question: 20



If there are m input lines and n output lines for a decoder that is used to uniquely address a byte-addressable 1 KB RAM, then the minimum value of $m + n$ is _____.

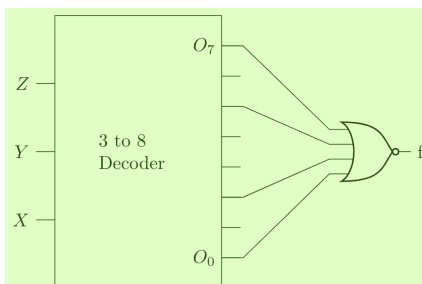
gatecse-2020 numerical-answers digital-logic decoder one-mark 6

Answer key

4.11.3 Decoder: GATE IT 2008 | Question: 9



What Boolean function does the circuit below realize?



- A. $xz + \bar{x}\bar{z}$ B. $x\bar{z} + \bar{x}z$
 C. $\bar{x}\bar{y} + yz$ D. $xy + \bar{y}\bar{z}$

gateit-2008 digital-logic circuit-output decoder normal

Answer key 11

4.12

Digital Circuits (7)

Practice Test: Test 1 (8Q) 12

4.12.1 Digital Circuits: GATE CSE 1996 | Question: 5



A logic network has two data inputs A and B , and two control inputs C_0 and C_1 . It implements the function F according to the following table.

C_1	C_0	F
0	0	$\overline{A+B}$
0	1	$A+B$
1	0	$A \oplus B$

Implement the circuit using one 4 to 1 Multiplexer, one 2-input Exclusive OR gate, one 2-input AND gate, one 2-input OR gate and one Inverter.

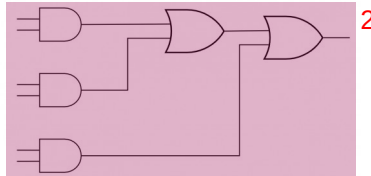
gate1996 digital-logic normal digital-circuits descriptive

Answer key

4.12.2 Digital Circuits: GATE CSE 2002 | Question: 7



- A. Express the function $f(x, y, z) = xy' + yz'$ with only one complement operation and one or more AND/OR operations. Draw the logic circuit implementing the expression obtained, using a single NOT gate and one or more AND/OR gates. 1
- B. Transform the following logic circuit (without expressing its switching function) into an equivalent logic circuit that employs only 6 NAND gates each with 2-inputs. 2



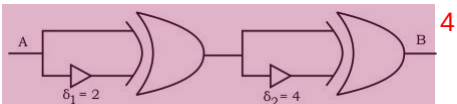
gatecse-2002 digital-logic normal descriptive digital-circuits

Answer key

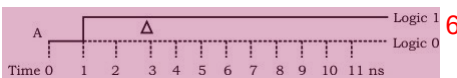
4.12.3 Digital Circuits: GATE CSE 2003 | Question: 47



Consider the following circuit composed of XOR gates and non-inverting buffers. 3



The non-inverting buffers have delays $\delta_1 = 2ns$ and $\delta_2 = 4ns$ as shown in the figure. Both XOR gates and all wires have zero delays. Assume that all gate inputs, outputs, and wires are stable at logic level 0 at time 0. If the following waveform is applied at input A , how many transition(s) (change of logic levels) occur(s) at B during the interval from 0 to 10 ns? 5



- A. 1 B. 2 C. 3 D. 4 7

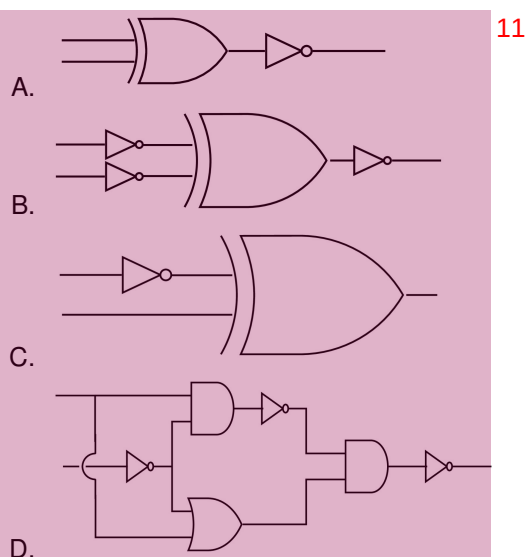
gatecse-2003 digital-logic digital-circuits 8

Answer key 9

4.12.4 Digital Circuits: GATE CSE 2011 | Question: 13



Which one of the following circuits is **NOT** equivalent to a 2-input *XNOR* (exclusive *NOR*) gate? 10



Answer key

4.12.5 Digital Circuits: GATE CSE 2013 | Question: 5



In the following truth table, $V = 1$ if and only if the input is valid.¹

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?³

- A. Priority encoder B. Decoder⁴
C. Multiplexer D. Demultiplexer

Answer key

4.12.6 Digital Circuits: GATE CSE 2014 | Set 3 | Question: 8



Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.⁵

```
f(x, a, b, y)
{
  if(x is 1) y = a;
  else y = b;
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?⁷

- A. Full adder B. Priority encoder C. Multiplexor D. Flip-flop⁸

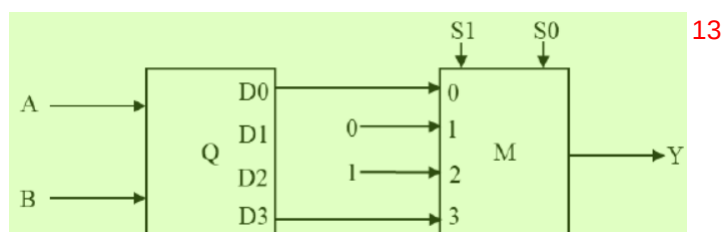
Answer key¹⁰

4.12.7 Digital Circuits: GATE CSE 2026 | Set 2 | Question: 49



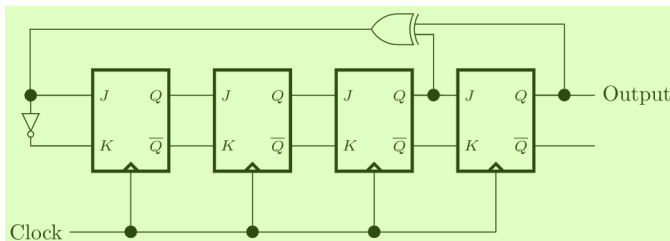
Consider the digital circuit shown below with two input lines A and B , two select lines S_0 and S_1 , and an output line Y . The blocks Q and M represent active high 2 : 4 decoder and 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce $Y = 1$ is _____ (answer in integer)¹¹

Note: One input combination is an instance of $[A \ B \ S_1 \ S_0]$.¹²



Answer key

4.13.1 Digital Counter: GATE CSE 1987 | Question: 1-III



The above circuit produces the output sequence: 2

- A. 1111 1111 0000 0000 B. 1111 0000 1111 0000 3
C. 1111 0001 0011 0101 D. 1010 1010 1010 1010

gate1987 digital-logic sequential-circuit flip-flop digital-counter

Answer key

4.13.2 Digital Counter: GATE CSE 1987 | Question: 10c



Give a minimal DFA that performs as a $\text{mod } -3$, 1's counter, i.e. outputs a 1 each time the number of 1's in the input sequence is a multiple of 3. 4

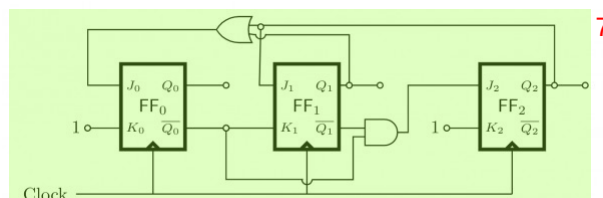
gate1987 digital-logic digital-counter descriptive 5

Answer key

4.13.3 Digital Counter: GATE CSE 1990 | Question: 5-c



For the synchronous counter shown in Fig.3, write the truth table of Q_0, Q_1 , and Q_2 after each pulse, starting from $Q_0 = Q_1 = Q_2 = 0$ and determine the counting sequence and also the modulus of the counter. 6



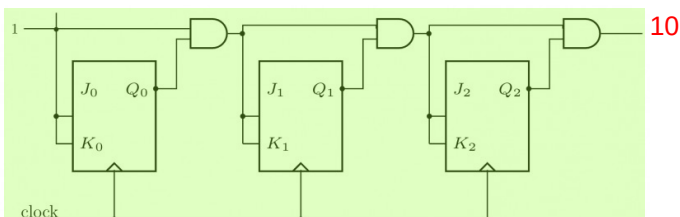
gate1990 descriptive digital-logic sequential-circuit flip-flop digital-counter 8

Answer key

4.13.4 Digital Counter: GATE CSE 1991 | Question: 5-c



Find the maximum clock frequency at which the counter in the figure below can be operated. Assume that the propagation delay through each flip flop and each AND gate is 10 ns . Also, assume that the setup time for the JK inputs of the flip flops is negligible. 9



gate1991 digital-logic sequential-circuit flip-flop digital-counter

Answer key

4.13.5 Digital Counter: GATE CSE 1994 | Question: 2-1



The number of flip-flops required to construct a binary modulo N counter is _____ 1

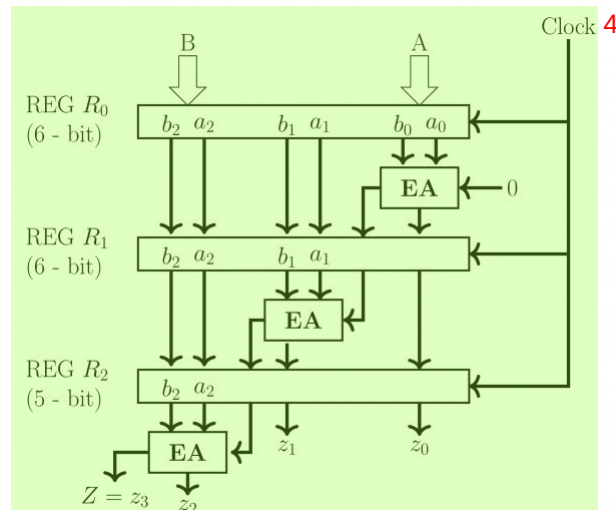
gate1994 digital-logic sequential-circuit flip-flop digital-counter fill-in-the-blanks 2

Answer key

4.13.6 Digital Counter: GATE CSE 2002 | Question: 8



Consider the following circuit. $A = a_2a_1a_0$ and $B = b_2b_1b_0$ are three bit binary numbers input to the circuit. The output is $Z = z_3z_2z_1z_0$. R_0 , R_1 and R_2 are registers with loading clock shown. The registers are loaded with their input data with the falling edge of a clock pulse (signal CLOCK shown) and appears as shown. The bits of input number A , B and the full adders are as shown in the circuit. Assume Clock period is greater than the settling time of all circuits. 3



a. For 8 clock pulses on the CLOCK terminal and the inputs A, B as shown, obtain the output Z (sequence of 4-bit values of Z). Assume initial contents of R_0, R_1 and R_2 as all zeros. 5

A	110	011	111	101	000	000	000	000
B	101	101	011	110	000	000	000	000
Clock No	1	2	3	4	5	6	7	8

b. What does the circuit implement? 7

gatecse-2002 digital-logic normal descriptive digital-counter 8

Answer key

4.13.7 Digital Counter: GATE CSE 2011 | Question: 15



The minimum number of D flip-flops needed to design a mod-258 counter is 9

A. 9 B. 8 C. 512 D. 258 10

gatecse-2011 digital-logic easy digital-counter 11

Answer key

4.13.8 Digital Counter: GATE CSE 2014 | Set 2 | Question: 7



Let $k = 2^n$. A circuit is built by giving the output of an n -bit binary counter as input to an n -to- 2^n bit decoder. This circuit is equivalent to a

A. k -bit binary up counter. B. k -bit binary down counter.
C. k -bit ring counter. D. k -bit Johnson counter.

gatecse-2014-set2 digital-logic normal digital-counter

Answer key

4.13.9 Digital Counter: GATE CSE 2015 | Set 1 | Question: 20



Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is ¹

- A. 0,1,3,7,15,14,12,8,0 B. 0,1,3,5,7,9,11,13,15,0 ²
C. 0,2,4,6,8,10,12,14,0 D. 0,8,12,14,15,7,3,1,0

gatecse-2015-set1 digital-logic digital-counter easy ³

Answer key

4.13.10 Digital Counter: GATE CSE 2015 | Set 2 | Question: 7



The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0, 0, 1, 1, 2, 2, 3, 3, 0, 0, ...) is ⁴_____.

gatecse-2015-set2 digital-logic digital-counter normal numerical-answers ⁵

Answer key

4.13.11 Digital Counter: GATE CSE 2016 | Set 1 | Question: 8



We want to design a synchronous counter that counts the sequence 0 – 1 – 0 – 2 – 0 – 3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is ⁶_____.

gatecse-2016-set1 digital-logic digital-counter flip-flop normal numerical-answers ⁷

Answer key

4.13.12 Digital Counter: GATE CSE 2017 | Set 2 | Question: 42



The next state table of a 2-bit saturating up-counter is given below. ⁹

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

¹⁰

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are ¹¹

- A. $T_1 = Q_1 Q_0$, $T_0 = \bar{Q}_1 \bar{Q}_0$ ¹²
B. $T_1 = \bar{Q}_1 Q_0$, $T_0 = \bar{Q}_1 + \bar{Q}_0$
C. $T_1 = Q_1 + Q_0$, $T_0 = \bar{Q}_1 \bar{Q}_0$
D. $T_1 = \bar{Q}_1 Q_0$, $T_0 = Q_1 + Q_0$

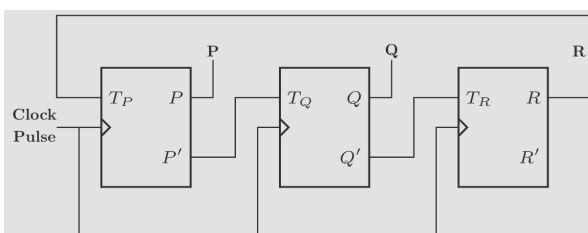
gatecse-2017-set2 digital-logic digital-counter

Answer key

4.13.13 Digital Counter: GATE CSE 2021 | Set 1 | Question: 28



Consider a 3-bit counter, designed using T flip-flops, as shown below:



Assuming the initial state of the counter given by PQR as 000, what are the next three states?

A. 011,101,000 B. 001,010,111 C. 011,101,111 D. 001,010,000 ¹

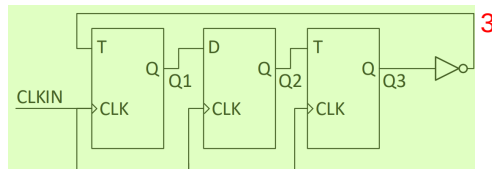
gatecse-2021-set1 digital-logic sequential-circuit digital-counter two-marks

Answer key

4.13.14 Digital Counter: GATE CSE 2023 | Question: 33



Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q₁, Q₂ and Q₃ have values 0, 1 and 1, respectively. ²



Which one of the given values of (Q₁, Q₂, Q₃) can NEVER be obtained with this digital circuit? ⁴

A. (0,0,1) B. (1,0,0) C. (1,0,1) D. (1,1,1) ⁵

gatecse-2023 digital-logic flip-flop two-marks digital-counter ⁶

Answer key ⁷

4.13.15 Digital Counter: GATE CSE 2026 | Set 1 | Question: 27



Consider a 2-bit saturating up/down counter that performs the saturating up count when the input P is 0, and the saturating down count when P is 1. The Next State table of the counter is as shown. The counter is built as a synchronous sequential circuit using D flip-flops. ⁸

Input	Curent State		Next State	
P	Q ₁	Q ₀	Q ₁ ⁺	Q ₀ ⁺
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

Which one of the following options corresponds to the expressions for the inputs of the D flip-flops, D_1 and D_0 ? ¹⁰

- A. $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = PQ_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$ ¹¹
 B. $D_1 = \bar{P}Q_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = \bar{P}Q_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$
 C. $D_1 = \bar{P}Q_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = \bar{P}Q_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$
 D. $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$ $D_0 = PQ_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$

gatecse-2026-set1 digital-logic digital-counter sequential-circuit flip-flop two-marks

Answer key

4.13.16 Digital Counter: GATE IT 2005 | Question: 11



How many pulses are needed to change the contents of a 8-bit up counter from 10101100 to 00100111 (rightmost bit is the LSB)?

A. 134 B. 133 C. 124 D. 123

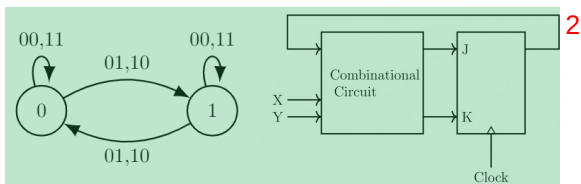
gateit-2005 digital-logic digital-counter normal

Answer key

4.13.17 Digital Counter: GATE IT 2008 | Question: 37



Consider the following state diagram and its realization by a JK flip flop 1



The combinational circuit generates J and K in terms of x, y and Q. 3
The Boolean expressions for J and K are :

- A. $\overline{x \oplus y}$ and $\overline{x \oplus y}$ B. $\overline{x \oplus y}$ and $x \oplus y$ 4
C. $x \oplus y$ and $\overline{x \oplus y}$ D. $x \oplus y$ and $x \oplus y$

gateit-2008 digital-logic boolean-algebra normal digital-counter 5

Answer key

4.13.18 Digital Counter: GATE1992-04-c



Design a 3-bit counter using D-flip flops such that not more than one flip-flop changes state between any two consecutive states. 6

gate1992 digital-logic sequential-circuit flip-flop digital-counter normal descriptive 7

Answer key

4.14 Dual Function (1)

4.14.1 Dual Function: GATE CSE 2014 | Set 2 | Question: 6



The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written as F^D is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

- A. 2^n B. 2^{n-1} C. 2^{2^n} D. $2^{2^{n-1}}$ 9

gatecse-2014-set2 digital-logic normal dual-function boolean-algebra 10

Answer key

4.15 Finite State Machines (4)

4.15.1 Finite State Machines: GATE CSE 1994 | Question: 3.3



State True or False with one line explanation 11

A FSM (Finite State Machine) can be designed to add two integers of any arbitrary length (arbitrary number of digits). 12

gate1994 digital-logic normal true-false finite-state-machines 13

Answer key

4.15.2 Finite State Machines: GATE CSE 1995 | Question: 2.23



A finite state machine with the following state table has a single input x and a single out z .

present state	next state, z	
	x=1	x=0
A	D,0	B,0
B	B,1	C,1
C	B,0	D,1
D	B,1	C,0

If the initial state is unknown, then the shortest input sequence to reach the final state C is: 1

- A. 01 B. 10 C. 101 D. 110 2

gate1995 digital-logic normal finite-state-machines 3

Answer key

4.15.3 Finite State Machines: GATE CSE 1996 | Question: 2.23



Consider the following state table for a sequential machine. The number of states in the minimized machine will be 4

		Input	
		0	1
Present State	A	D,0	B,1
	B	A,0	C,1
	C	A,0	B,1
	D	A,1	C,1
		Next state, Output	

- A. 4 B. 3 C. 2 D. 1 6

gate1996 normal digital-logic finite-state-machines 7

Answer key 8

4.15.4 Finite State Machines: GATE CSE 2009 | Question: 27



Given the following state table of an FSM with two states A and B , one input and one output. 9

PRESENT STATE A	PRESENT STATE B	Input	Next State A	Next State B	Output	10
0	0	0	0	0	1	
0	1	0	1	0	0	
1	0	0	0	1	0	
1	1	0	1	0	0	
0	0	1	0	1	0	
0	1	1	0	0	1	
1	0	1	0	1	1	
1	1	1	0	0	1	

If the initial state is $A = 0, B = 0$ what is the minimum length of an input string which will take the machine to the state $A = 0, B = 1$ with *output* = 1. 11

- A. 3 B. 4 C. 5 D. 6 12

gatecse-2009 digital-logic normal finite-state-machines 13

Answer key

4.16

Fixed Point Representation (2)

4.16.1 Fixed Point Representation: GATE CSE 2017 | Set 1 | Question: 7



The n -bit fixed-point representation of an unsigned real number X uses f bits for the fraction part. Let $i = n - f$. The range of decimal values for X in this representation is

- A. 2^{-f} to 2^i B. 2^{-f} to $(2^i - 2^{-f})$
 C. 0 to 2^i D. 0 to $(2^i - 2^{-f})$

gatecse-2017-set1 digital-logic number-representation fixed-point-representation

Answer key

4.16.2 Fixed Point Representation: GATE CSE 2018 | Question: 33



Consider the unsigned 8-bit fixed point binary number representation, below, ¹

$$b_7 b_6 b_5 b_4 b_3 \cdot b_2 b_1 b_0$$

where the position of the primary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation: ³

- i. 31.500 ⁴
- ii. 0.875
- iii. 12.100
- iv. 3.001

Which one of the following statements is true? ⁵

- A. None of i, ii, iii, iv can be exactly represented ⁶
- B. Only ii cannot be exactly represented
- C. Only iii and iv cannot be exactly represented
- D. Only i and ii cannot be exactly represented

gatecse-2018 digital-logic number-representation fixed-point-representation normal two-marks ⁷

Answer key

4.17

Flip Flop (7)

Practice Tests: Test 1 (15Q) Test 2 (6Q) ⁸

4.17.1 Flip Flop: GATE CSE 2001 | Question: 11



A sequential circuit takes an input stream of 0's and 1's and produces an output stream of 0's and 1's. Initially it replicates the input on its output until two consecutive 0's are encountered on the input. From then onward, it produces an output stream, which is the bit-wise complement of input stream until it encounters two consecutive 1's, whereupon the process repeats. An example input and output stream is shown below. ⁹

The input stream: 101100|01001011|011 ¹⁰

The desired output: 101100|10110100|011

J-K master-slave flip-flops are to be used to design the circuit. ¹¹

- a. Give the state transition diagram ¹²
- b. Give the minimized sum-of-product expression for J and K inputs of one of its state flip-flops

gatecse-2001 digital-logic normal descriptive flip-flop

Answer key

4.17.2 Flip Flop: GATE CSE 2004 | Question: 18, ISRO2007-31



In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

- A. $Q = 0, Q' = 1$
- B. $Q = 1, Q' = 0$
- C. $Q = 1, Q' = 1$
- D. Indeterminate states

gatecse-2004 digital-logic easy isro2007 flip-flop

Answer key

4.17.3 Flip Flop: GATE CSE 2015 | Set 1 | Question: 37



A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q

output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that $J = K = 1$ is the toggle mode and $J = K = 0$ is the state holding mode of the JK flip-flops. Both the flip-flops have non-zero propagation delays.

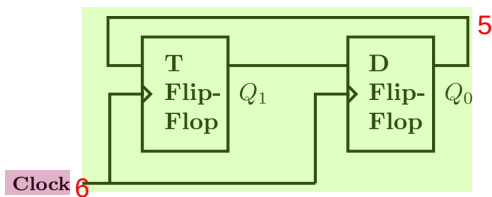
- A. 0110110...
 B. 0100100...
 C. 011101110...
 D. 011001100...

gatecse-2015-set1 digital-logic flip-flop normal 3

Answer key

4.17.4 Flip Flop: GATE CSE 2017 | Set 1 | Question: 33

Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

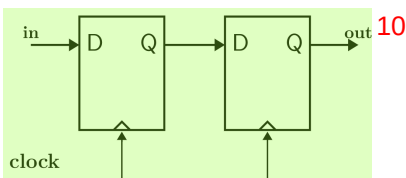
- A. Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively.
 B. Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively.
 C. Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively.
 D. Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively.

gatecse-2017-set1 digital-logic flip-flop normal

Answer key

4.17.5 Flip Flop: GATE CSE 2018 | Question: 22

Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



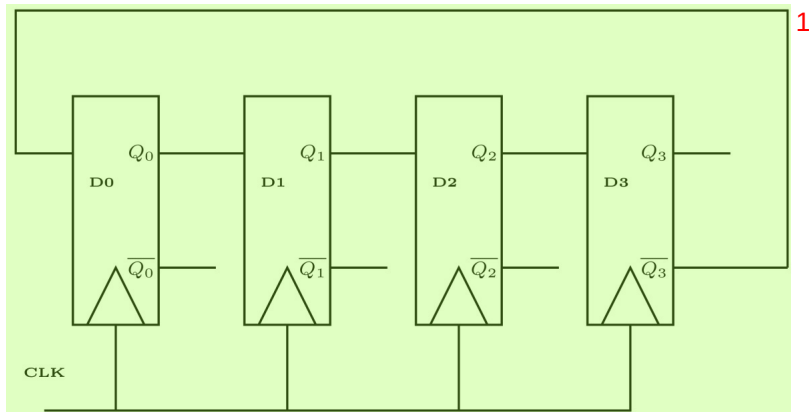
The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is ____

gatecse-2018 digital-logic flip-flop numerical-answers normal one-mark 12

Answer key

4.17.6 Flip Flop: GATE CSE 2025 | Set 1 | Question: 50

Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____. (Answer in integer)



1

gatecse2025-set1 digital-logic flip-flop sequential-circuit numerical-answers two-marks

Answer key

4.17.7 Flip Flop: GATE IT 2007 | Question: 7

Which of the following input sequences for a cross-coupled $R - S$ flip-flop realized with two $NAND$ gates may lead to an oscillation?

- A. 11,00 B. 01,10 C. 10,01 D. 00,11 ³

gateit-2007 digital-logic normal flip-flop ⁴

Answer key

4.18 Floating Point Representation (8)

Practice Test: Test 1 (11Q)

4.18.1 Floating Point Representation: GATE CSE 1987 | Question: 1-vii

The exponent of a floating-point number is represented in excess- N code so that: ⁶

- A. The dynamic range is large. B. The precision is high. ⁷
C. The smallest number is represented by all zeros. D. Overflow is avoided.

gate1987 digital-logic number-representation floating-point-representation ⁸

Answer key

4.18.2 Floating Point Representation: GATE CSE 1989 | Question: 1-vi

Consider an excess -50 representation for floating point numbers with 4 BCD digit mantissa and 2 BCD digit exponent in normalised form. The minimum and maximum positive numbers that can be represented are _____ and _____ respectively. ⁹

descriptive gate1989 digital-logic number-representation floating-point-representation ¹⁰

Answer key

4.18.3 Floating Point Representation: GATE CSE 1990 | Question: 1-iv-a

A 32-bit floating-point number is represented by a 7-bit signed exponent, and a 24-bit fractional mantissa. The base of the scale factor is 16,
The range of the exponent is _____

gate1990 digital-logic number-representation floating-point-representation fill-in-the-blanks

Answer key

4.18.4 Floating Point Representation: GATE CSE 1990 | Question: 1-iv-b

A 32-bit floating-point number is represented by a 7-bit signed exponent, and a 24-bit fractional mantissa.

The base of the scale factor is 16,
The range of the exponent is _____, if the scale factor is represented in excess-64 format.

gate1990 digital-logic number-representation floating-point-representation fill-in-the-blanks

Answer key

4.18.5 Floating Point Representation: GATE CSE 1997 | Question: 72



Following floating point number format is given

f is a fraction represented by a 6-bit mantissa (includes sign bit) in sign magnitude form, e is a 4-bit exponent (includes sign bit) in sign magnitude form and $n = (f, e) = f \cdot 2^e$ is a floating point number. Let $A = 54.75$ in decimal and $B = 9.75$ in decimal

- Represent A and B as floating point numbers in the above format.
- Show the steps involved in floating point addition of A and B .
- What is the percentage error (up to one position beyond decimal point) in the addition operation in (b)?

gate1997 digital-logic floating-point-representation normal descriptive

Answer key

4.18.6 Floating Point Representation: GATE CSE 2003 | Question: 43



The following is a scheme for floating point number representation using 16 bits.

Bit Position	15	14		9	8		0
	s	e			m		
	Sign	Exponent			Mantissa		

Let s, e , and m be the numbers represented in binary in the sign, exponent, and mantissa fields respectively. Then the floating point number represented is:

$$\begin{cases} (-1)^s (1 + m \times 2^{-9}) 2^{e-31}, & \text{if the exponent} \neq 11111 \\ 0, & \text{otherwise} \end{cases}$$

What is the maximum difference between two successive real numbers representable in this system?

- A. 2^{-40} B. 2^{-9} C. 2^{22} D. 2^{31}

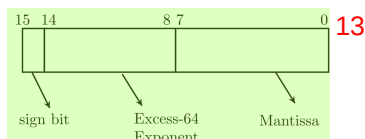
gatecse-2003 digital-logic number-representation floating-point-representation normal

Answer key

4.18.7 Floating Point Representation: GATE CSE 2005 | Question: 85-a



Consider the following floating-point format.



Mantissa is a pure fraction in sign-magnitude form.

The decimal number 0.239×2^{13} has the following hexadecimal representation (without normalization and rounding off):

- A. 0D 24 B. 0D 4D C. 4D 0D D. 4D 3D

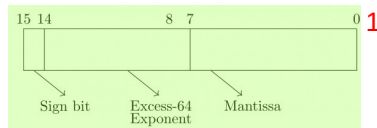
gatecse-2005 digital-logic number-representation floating-point-representation normal

Answer key

4.18.8 Floating Point Representation: GATE CSE 2005 | Question: 85-b



Consider the following floating-point format.



Mantissa is a pure fraction in sign-magnitude form. **2**

The normalized representation for the above format is specified as follows. The mantissa has an implicit 1 preceding the binary (radix) point. Assume that only 0's are padded in while shifting a field. **3**

The normalized representation of the above number (0.239×2^{13}) is: **4**

- A. 0A 20 B. 11 34 C. 49 D0 D. 4A E8 **5**

gatecse-2005 digital-logic number-representation floating-point-representation normal **6**

Answer key

4.19 Functional Completeness (7)

Practice Test: Test 1 (5Q)

4.19.1 Functional Completeness: GATE CSE 1989 | Question: 4-iii



Show that {NOR} is a functionally complete set of Boolean operations. **7**

gate1989 descriptive digital-logic functional-completeness **8**

Answer key

4.19.2 Functional Completeness: GATE CSE 1992 | Question: 02-ii



All digital circuits can be realized using only **9**

- A. Ex-OR gates B. Multiplexers C. Half adders D. OR gates **10**

gate1992 normal digital-logic digital-circuits multiple-selects functional-completeness combinational-circuit **11**

Answer key

4.19.3 Functional Completeness: GATE CSE 1993 | Question: 9



Assume that only half adders are available in your laboratory. Show that any binary function can be implemented using half adders only. **12**

gate1993 digital-logic combinational-circuit adder descriptive functional-completeness **13**

Answer key

4.19.4 Functional Completeness: GATE CSE 1998 | Question: 5



The implication gate, shown below has two inputs (x and y); the output is 1 except when $x = 1$ and $y = 0$, realize $f = \bar{x}y + x\bar{y}$ using only four implication gates. **14**



Show that the implication gate is functionally complete. **16**

gate1998 digital-logic functional-completeness descriptive **17**

Answer key

4.19.5 Functional Completeness: GATE CSE 1999 | Question: 2.9



Which of the following sets of component(s) is/are sufficient to implement any arbitrary Boolean function?

- A. XOR gates, NOT gates
- B. 2 to 1 multiplexers
- C. AND gates, XOR gates
- D. Three-input gates that output $(A \cdot B) + C$ for the inputs A, B and C .

1

gate1999 digital-logic normal functional-completeness multiple-selects 2

Answer key

4.19.6 Functional Completeness: GATE CSE 2015 | Set 1 | Question: 39

Consider the operations

$$f(X, Y, Z) = X'YZ + XY' + Y'Z' \text{ and } g(X, Y, Z) = X'YZ + X'YZ' + XY$$

Which one of the following is correct?

3

- A. Both $\{f\}$ and $\{g\}$ are functionally complete
- B. Only $\{f\}$ is functionally complete
- C. Only $\{g\}$ is functionally complete
- D. Neither $\{f\}$ nor $\{g\}$ is functionally complete

4

gatecse-2015-set1 boolean-algebra difficult functional-completeness 5

Answer key

4.19.7 Functional Completeness: GATE IT 2008 | Question: 1

A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?

- A. EX-NOR
- B. implication, negation
- C. OR, negation
- D. NAND

gateit-2008 digital-logic easy functional-completeness 8

Answer key

4.20

IEEE Representation (14)

Practice Tests: Test 1 (15Q) Test 2 (6Q)

4.20.1 IEEE Representation: GATE CSE 2008 | Question: 4

In the IEEE floating point representation the hexadecimal value $0x00000000$ corresponds to

- A. The normalized value 2^{-127}
- B. The normalized value 2^{-126}
- C. The normalized value $+0$
- D. The special value $+0$

11

gatecse-2008 digital-logic floating-point-representation ieee-representation easy 12

Answer key

4.20.2 IEEE Representation: GATE CSE 2012 | Question: 7

The decimal value 0.5 in IEEE single precision floating point representation has

- A. fraction bits of $000 \dots 000$ and exponent value of 0
- B. fraction bits of $000 \dots 000$ and exponent value of -1
- C. fraction bits of $100 \dots 000$ and exponent value of 0
- D. no exact representation

gatecse-2012 digital-logic normal number-representation ieee-representation

Answer key

4.20.3 IEEE Representation: GATE CSE 2014 | Set 2 | Question: 45



The value of a float type variable is represented using the single-precision 32-bit floating point format of IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for the mantissa. A float type variable X is assigned the decimal value of -14.25 . The representation of X in hexadecimal notation is

- A. C1640000H B. 416C0000H C. 41640000H D. C16C0000H ²

gatecse-2014-set2 digital-logic number-representation normal ieee-representation ³

Answer key

4.20.4 IEEE Representation: GATE CSE 2017 | Set 2 | Question: 12



Given the following binary number in 32-bit (single precision) IEEE-754 format : ⁴

00111110011011010000000000000000 ⁵

The decimal value closest to this floating-point number is : ⁶

- A. $1.45 * 10^1$ B. $1.45 * 10^{-1}$ C. $2.27 * 10^{-1}$ D. $2.27 * 10^1$ ⁷

gatecse-2017-set2 digital-logic ieee-representation number-representation floating-point-representation ⁸

Answer key

4.20.5 IEEE Representation: GATE CSE 2020 | Question: 29



Consider three registers $R1$, $R2$, and $R3$ that store numbers in IEEE-754 single precision floating point format. Assume that $R1$ and $R2$ contain the values (in hexadecimal notation) $0x42200000$ and $0xC1200000$, respectively.

If $R3 = \frac{R1}{R2}$, what is the value stored in $R3$? ¹⁰

- A. $0x40800000$ B. $0xC0800000$ C. $0x83400000$ D. $0xC8500000$ ¹¹

gatecse-2020 digital-logic ieee-representation floating-point-representation two-marks ¹²

Answer key

4.20.6 IEEE Representation: GATE CSE 2021 | Set 1 | Question: 24



Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. ¹³

$S : 1$ $E : 10000001$ $F : 111100000000000000000000$ ¹⁵

Here S , E and F denote the sign, exponent, and fraction components of the floating point representation. ¹⁶

The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.

gatecse-2021-set1 digital-logic number-representation ieee-representation numerical-answers one-mark floating-point-representation ¹⁷

Answer key

4.20.7 IEEE Representation: GATE CSE 2021 | Set 2 | Question: 4



The format of the single-precision floating point representation of a real number as per the IEEE 754 standard is as follows:

sign	exponent	mantissa
------	----------	----------

Which one of the following choices is correct with respect to the *smallest* normalized positive number represented using the standard?

- A. exponent = 00000000 and mantissa = 000000000000000000000000
 B. exponent = 00000000 and mantissa = 000000000000000000000001
 C. exponent = 00000001 and mantissa = 000000000000000000000000

D. exponent = 00000001 and mantissa = 000000000000000000000001

gatecse-2021-set2	digital-logic	number-representation	ieee-representation	one-mark	floating-point-representation
-------------------	---------------	-----------------------	---------------------	----------	-------------------------------

Answer key

4.20.8 IEEE Representation: GATE CSE 2022 | Question: 31



Consider three floating point numbers A , B and C stored in registers R_A , R_B and R_C , respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows.

$R_A = 0xC1400000$	$R_B = 0x42100000$	$R_C = 0x41400000$
--------------------	--------------------	--------------------

Which one of the following is FALSE? 3

- A. $A + C = 0$ B. $C = A + B$ C. $B = 3C$ D. $(B - C) > 0$ 4

gatecse-2022	digital-logic	ieee-representation	number-representation	two-marks	floating-point-representation	5
--------------	---------------	---------------------	-----------------------	-----------	-------------------------------	---

Answer key 

4.20.9 IEEE Representation: GATE CSE 2023 | Question: 35



Consider the **IEEE-754** single precision floating point numbers $P = 0xC1800000$ and $Q = 0x3F5C2EF4$. Which one of the following corresponds to the product of these numbers (i.e., $P \times Q$), represented in the **IEEE-754** single precision format?

- A. 0x404C2EF4 B. 0x405C2EF4 C. 0xC15C2EF4 D. 0xC14C2EF4 7

gatecse-2023 digital-logic number-representation ieee-representation two-marks floating-point-representation 8

Answer key 

4.20.10 IEEE Representation: GATE CSE 2024 | Set 2 | Question: 4



The format of a single-precision floating-point number as per the IEEE 754 standard is: 9

Sign	Exponent	Mantissa
(1 bit)	(8 bits)	(23 bits)

Choose the largest floating-point number among the following options. 11

A.	Sign	Exponent	Mantissa
	0	01111111	111111111111111111111111

B.	Sign	Exponent	Mantissa
	0	11111110	111111111111111111111111

C.	Sign	Exponent	Mantissa
	0	11111111	111111111111111111111111

D.	Sign	Exponent	Mantissa
	0	01111111	000000000000000000000000

gatecse-2024-set2 digital-logic number-representation ieee-representation floating-point-representation one-mark

Answer key

4.20.11 IEEE Representation: GATE CSE 2025 | Set 2 | Question: 39



Three floating point numbers X , Y , and Z are stored in three registers R_X , R_Y , and R_Z , respectively in *IEEE 754* single precision format as given below in hexadecimal:

$$R_X = 0xC1100000, R_Y = 0x40C00000, \text{ and } R_Z = 0x41400000$$

Which of the following option(s) is/are CORRECT? 1

A. $4(X + Y) + Z = 0$ B. $2Y - Z = 0$ C. $4X + 3Z = 0$ D. $X + Y + Z = 0$ 2

gatecse2025-set2 digital-logic number-representation ieee-representation multiple-selects two-marks 3

Answer key

4.20.12 IEEE Representation: GATE CSE 2026 | Set 1 | Question: 26

Consider the real valued variables X, Y and Z represented using the IEEE 754 singleprecision floating point format. The binary representations of X and Y in hexadecimal notation are as follows: 4

$X : 35C00000$ $Y : 34A00000$

Let $Z = X + Y$. 5

Which one of the following is the binary representation of Z , in hexadecimal notation? 6

A. $35C80000$ B. $35CC0000$ C. $35E80000$ D. $35EC0000$ 7

gatecse-2026-set1 digital-logic ieee-representation floating-point-representation two-marks 8

Answer key

4.20.13 IEEE Representation: GATE CSE 2026 | Set 2 | Question: 24

The 32-bit IEEE 754 single precision representation of a number is $0xC2710000$. The number in decimal representation is _____. (rounded off to two decimal places) 9

gatecse-2026-set2 digital-logic ieee-representation numerical-answers one-mark 10

Answer key

4.20.14 IEEE Representation: GATE IT 2008 | Question: 7

The following bit pattern represents a floating point number in IEEE 754 single precision format 11

1 10000011 101000000000000000000000

The value of the number in decimal form is

A. -10 B. -13 C. -26 D. None of the above 12

gateit-2008 digital-logic ieee-representation number-representation floating-point-representation normal 13

Answer key 14

4.21 K Map (17)

 Practice Tests: Test 1 (15Q) Test 2 (5Q) Weekly Quiz 1 (15Q) Weekly Quiz 2 (15Q)

4.21.1 K Map: GATE CSE 1987 | Question: 16-a

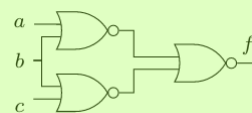
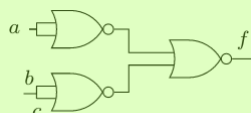
A Boolean function f is to be realized only by NOR gates. Its K -map is given below: 15

c \ ab	00	01	11	10
	0	0	1	1
1	0	1	1	1

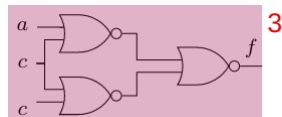
 16

The realization is 17

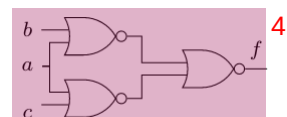
A. B. 18



C.1



D.2



gate1987 digital-logic k-map

Answer key

4.21.2 K Map: GATE CSE 1988 | Question: 3a-b



		BC						B	
		00	01	11	10			0	1
A	0	1		1	1	A	0	$\bar{C}$	1
	1		1	1				C	C

The Karnaugh map of a function of (A, B, C) is shown on the left hand side of the above figure.

The reduced form of the same map is shown on the right hand side, in which the variable C is entered in the map itself. Discuss,

- The methodology by which the reduced map has been derived and
- the rules (or steps) by which the boolean function can be derived from the entries in the reduced map.

gate1988 descriptive digital-logic k-map

Answer key

4.21.3 K Map: GATE CSE 1992 | Question: 01-i



The Boolean function in sum of products form where K-map is given below (figure) is _____

		C	
		0	1
B	0	1	0
	1	$\bar{A}$	A

gate1992 digital-logic k-map normal fill-in-the-blanks

Answer key

4.21.4 K Map: GATE CSE 1995 | Question: 15-b



What is the equivalent minimal Boolean expression (in sum of products form) for the Karnaugh map given below?

		AB			
		00	01	11	10
CD	00	1			1
	01		1	1	
	11		1	1	
	10	1			1

gate1995 digital-logic boolean-algebra k-map normal descriptive

Answer key

4.21.5 K Map: GATE CSE 1996 | Question: 2.24



What is the equivalent Boolean expression in product-of-sums form for the Karnaugh map given in Fig 1

CD \ AB	00	01	11	10
	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

- A. $B\bar{D} + \bar{B}D$ B. $(B + \bar{C} + D)(\bar{B} + C + \bar{D})$
 C. $(B + D)(\bar{B} + \bar{D})$ D. $(B + \bar{D})(\bar{B} + D)$

gate1996 digital-logic k-map easy 4

Answer key 5

4.21.6 K Map: GATE CSE 1998 | Question: 2.7



The function represented by the Karnaugh map given below is

A \ BC	00	01	10	11
	00	01	10	11
0	1	0	0	1
1	1	0	0	1

- A. $A.B$ B. $AB + BC + CA$ C. $\overline{B \oplus C}$ D. $A.BC$

gate1998 digital-logic k-map normal 9

Answer key 10

4.21.7 K Map: GATE CSE 1999 | Question: 1.8



Which of the following functions implements the Karnaugh map shown below?

AB \ CD	00	01	11	10
	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

- A. $\bar{A}B + CD$ B. $D(C + A)$
 C. $AD + \bar{A}B$ D. $(C + D)(\bar{C} + D) + (A + B)$

gate1999 digital-logic k-map easy 14

Answer key 15

4.21.8 K Map: GATE CSE 2000 | Question: 2.11



Which functions does NOT implement the Karnaugh map given below?

xy \ wz	00	01	11	10
	00	01	11	10
00	0	X	0	0
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

- A. $(w + x)y$ B. $xy + yw$

C. $(w+x)(\bar{w}+y)(\bar{x}+y)$ 1

D. None of the above 2

gatecse-2000 digital-logic k-map normal

Answer key 3

4.21.9 K Map: GATE CSE 2001 | Question: 1.11

Given the following karnaugh map, which one of the following represents the minimal Sum-Of-Products of the map? 4

$wx \backslash yz$	00	01	11	10
00	0	X	0	X
01	X	1	X	1
11	0	X	1	0
10	0	1	X	0

5

A. $XY + Y'Z$

B. $WX'Y' + XY + XZ$

C. $W'X + Y'Z + XY$

D. $XZ + Y$ 6

gatecse-2001 k-map digital-logic normal

Answer key 8

4.21.10 K Map: GATE CSE 2002 | Question: 1.12

Minimum sum of product expression for $f(w, x, y, z)$ shown in Karnaugh-map below 9

$wx \backslash yz$	00	01	11	10
00	0	1	1	0
01	X	0	0	1
11	X	0	0	1
10	0	1	1	X

10

A. $xz + y'z$

B. $xz' + zx'$

C. $x'y + zx'$

D. None of the above 11

gatecse-2002 digital-logic k-map normal

Answer key 13

4.21.11 K Map: GATE CSE 2003 | Question: 45

The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(xy + xz')$ is 4. What are the minimum possible literal counts of the product-of-sum and sum-of-product representations respectively of the function given by the following Karnaugh map? Here, X denotes "don't care" 14

$zw \backslash xy$	00	01	11	10
00	X	1	0	1
01	0	1	X	0
11	1	X	X	0
10	X	0	0	X

15

A. (11,9)

B. (9,13)

C. (9,10)

D. (11,11) 16

gatecse-2003 digital-logic k-map normal

Answer key

4.21.12 K Map: GATE CSE 2008 | Question: 5



In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

ab \ cd	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

- A. $\bar{b}. \bar{d} + \bar{a}. \bar{d}$ B. $\bar{a}. \bar{b} + \bar{b}. \bar{d} + \bar{a}. b. \bar{d}$
 C. $\bar{b}. \bar{d} + \bar{a}. b. \bar{d}$ D. $\bar{a}. \bar{b} + \bar{b}. \bar{d} + \bar{a}. \bar{d}$

gatescse-2008 digital-logic k-map easy 4

Answer key 5

4.21.13 K Map: GATE CSE 2012 | Question: 30



What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term

ab \ cd	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- A. $\bar{b}\bar{d}$ B. $\bar{b}\bar{d} + \bar{b}\bar{c}$
 C. $\bar{b}\bar{d} + a\bar{b}\bar{c}d$ D. $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

gatescse-2012 digital-logic k-map easy 9

Answer key 10

4.21.14 K Map: GATE CSE 2025 | Set 2 | Question: 33



Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$:

wx \ yz	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

Which one or more of the following Boolean expression(s) represent(s) F ?

- A. $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$
 B. $\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$
 C. $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}y\bar{z} + xz$
 D. $\bar{x}\bar{z} + xz$

gatescse2025-set2 digital-logic k-map multiple-selects two-marks

Answer key

4.21.15 K Map: GATE IT 2006 | Question: 35

The boolean function for a combinational circuit with four inputs is represented by the following Karnaugh map.

RS \ PQ	00	01	11	10
	00	01	11	10
00	1	0	0	1
01	0	0	1	1
11	1	1	1	0
10	1	0	0	1

Which of the product terms given below is an essential prime implicant of the function? 3

- A. QRS B. PQS C. PQ'S' D. Q'S' 4

gateit-2006 digital-logic k-map normal 5

Answer key

4.21.16 K Map: GATE IT 2007 | Question: 78

Consider the following expression

$$a\bar{d} + \bar{a}\bar{c} + b\bar{c}d$$

Which of the following Karnaugh Maps correctly represents the expression? 6

A.

cd \ ab	00	01	11	10
00	X	X		
01	X	X		
11	X	X		X
10	X			X

B.

cd \ ab	00	01	11	10
00	X	X		
01	X			
11	X	X		X
10	X	X		X

C.

cd \ ab	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X			X

D.

cd \ ab	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X		X	X

 7

gateit-2007 digital-logic k-map normal 8

Answer key

4.21.17 K Map: GATE IT 2007 | Question: 79

Consider the following expression 9

$$a\bar{d} + \bar{a}\bar{c} + b\bar{c}d$$

Which of the following expressions does not correspond to the Karnaugh Map obtained for the given expression? 10

- A. $\bar{c}\bar{d} + a\bar{d} + ab\bar{c} + \bar{a}\bar{c}d$ 11
 B. $\bar{a}\bar{c} + \bar{c}\bar{d} + a\bar{d} + ab\bar{c}d$
 C. $\bar{a}\bar{c} + a\bar{d} + ab\bar{c} + \bar{c}d$
 D. $\bar{b}\bar{c}\bar{d} + ac\bar{d} + \bar{a}\bar{c} + ab\bar{c}$

gateit-2007 digital-logic k-map normal 12

Answer key

4.22

Little Endian Big Endian (1)

4.22.1 Little Endian Big Endian: GATE CSE 2021 | Set 2 | Question: 44

If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?

- A. 0x6665 B. 0x0001 C. 0x4243 D. 0x0100

Answer key

4.23

Memory Interfacing (5)

Practice Test: Test 1 (5Q)

4.23.1 Memory Interfacing: GATE CSE 1995 | Question: 2.2



The capacity of a memory unit is defined by the number of words multiplied by the number of bits/word. How many separate address and data lines are needed for a memory of $4K \times 16$?

- A. 10 address, 16 data lines B. 11 address, 8 data lines
C. 12 address, 16 data lines D. 12 address, 12 data lines

gate1995 digital-logic memory-interfacing normal 4

Answer key

4.23.2 Memory Interfacing: GATE CSE 2009 | Question: 7, ISRO2015-3



How many $32K \times 1$ RAM chips are needed to provide a memory capacity of $256K$ bytes?

- A. 8 B. 32 C. 64 D. 128

gatecse-2009 digital-logic memory-interfacing easy isro2015 7

Answer key

4.23.3 Memory Interfacing: GATE CSE 2010 | Question: 7



The main memory unit with a capacity of 4 megabytes is built using $1M \times 1$ -bit DRAM chips. Each DRAM chip has $1K$ rows of cells with $1K$ cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is

- A. 100 nanoseconds B. 100×2^{10} nanoseconds
C. 100×2^{20} nanoseconds D. 3200×2^{20} nanoseconds

gatecse-2010 digital-logic memory-interfacing normal 11

Answer key

4.23.4 Memory Interfacing: GATE CSE 2013 | Question: 46



A RAM chip has a capacity of 1024 words of 8 bits each ($1K \times 8$). The number of 2×4 decoders with enable line needed to construct a $16K \times 16$ RAM from $1K \times 8$ RAM is

- (A) 4 (B) 5 (C) 6 (D) 7

gatecse-2013 digital-logic normal memory-interfacing 14

Answer key

4.23.5 Memory Interfacing: GATE IT 2005 | Question: 9



A dynamic RAM has a memory cycle time of 64 nsec. It has to be refreshed 100 times per msec and each refresh takes 100 nsec. What percentage of the memory cycle time is used for refreshing?

- A. 10 B. 6.4 C. 1 D. 0.64

gateit-2005 digital-logic memory-interfacing normal 18

Answer key 19

4.24

Min No Gates (6)

Practice Tests: Test 1 (15Q) Test 2 (7Q)

4.24.1 Min No Gates: GATE CSE 1995 | Question: 15-a



Implement a circuit having the following output expression using an inverter and a nand gate ¹

$$Z = \overline{A} + \overline{B} + C$$

gate1995 digital-logic normal descriptive min-no-gates digital-circuits boolean-algebra ³

Answer key ⁴

4.24.2 Min No Gates: GATE CSE 2000 | Question: 9



Design a logic circuit to convert a single digit BCD number to the number modulo six as follows (Do not detect illegal input):

- Write the truth table for all bits. Label the input bits $I_1, I_2, \dots$ with I_1 as the least significant bit. Label the output bits $R_1, R_2, \dots$ with R_1 as the least significant bit. Use 1 to signify truth. ⁶
- Draw one circuit for each output bit using, **altogether**, two two-input AND gates, one two-input OR gate and two NOT gates.

gatecse-2000 digital-logic min-no-gates descriptive ⁷

Answer key

4.24.3 Min No Gates: GATE CSE 2004 | Question: 58



A circuit outputs a digit in the form of 4 bits. 0 is represented by 0000, 1 by 0001, ..., 9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and outputs 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?

- A. 2 B. 3 C. 4 D. 5 ⁹

gatecse-2004 digital-logic normal min-no-gates

Answer key

4.24.4 Min No Gates: GATE CSE 2009 | Question: 6



What is the minimum number of gates required to implement the Boolean function $(AB+C)$ if we have to use only 2-input NOR gates? ¹⁰

- A. 2 B. 3 C. 4 D. 5 ¹²

gatecse-2009 digital-logic min-no-gates normal

Answer key

4.24.5 Min No Gates: GATE CSE 2017 | Set 1 | Question: 21



Consider the Karnaugh map given below, where X represents "don't care" and blank represents 0. ¹³

ba \ dc	00	01	11	10
00		X	X	
01	1			X
11	1			1
10		X	X	

¹⁴

Assume for all inputs (a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____. ¹⁵

gatecse-2017-set1 digital-logic k-map numerical-answers normal min-no-gates

Answer key

4.24.6 Min No Gates: GATE IT 2004 | Question: 8



What is the minimum number of NAND gates required to implement a 2-input EXCLUSIVE-OR function without using any other logic gate?

- A. 2 B. 4 C. 5 D. 6

gateit-2004 digital-logic min-no-gates normal 3

Answer key

4.25

Min Products of Sum Form (2)

4.25.1 Min Products of Sum Form: GATE CSE 1990 | Question: 5-a



Find the minimum product of sums of the following expression

$$f = ABC + \bar{A} \bar{B} \bar{C}$$

gate1990 digital-logic boolean-algebra min-products-of-sum-form canonical-normal-form descriptive 6

Answer key

4.25.2 Min Products of Sum Form: GATE CSE 2017 | Set 2 | Question: 28



Given $f(w, x, y, z) = \sum_m(0, 1, 2, 3, 7, 8, 10) + \sum_d(5, 6, 11, 15)$; where d represents the 'don't-care' condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$?

- A. $f = (\bar{w} + \bar{z})(\bar{x} + z)$ B. $f = (\bar{w} + z)(x + z)$
 C. $f = (w + z)(\bar{x} + z)$ D. $f = (w + \bar{z})(\bar{x} + z)$

gatecse-2017-set2 digital-logic min-products-of-sum-form 10

Answer key

4.26

Min Sum of Products Form (16)

Practice Tests: Test 1 (15Q) Test 2 (1Q) 14

4.26.1 Min Sum of Products Form: GATE CSE 1991 | Question: 5-b



Find the minimum sum of products form of the logic function $f(A, B, C, D) = \sum_m(0, 2, 8, 10, 15) + \sum_d(3, 11, 12, 14)$ where m and d represent minterm and don't care term respectively.

gate1991 digital-logic boolean-algebra min-sum-of-products-form descriptive 17

Answer key

4.26.2 Min Sum of Products Form: GATE CSE 1997 | Question: 71



Let $f = (\bar{w} + y)(\bar{x} + y)(w + \bar{x} + z)(\bar{w} + z)(\bar{x} + z)$

- A. Express f as the minimal sum of products. Write only the answer.
 B. If the output line is stuck at 0, for how many input combinations will the value of f be correct?

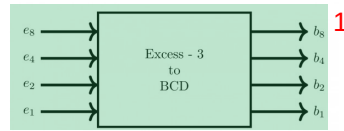
gate1997 digital-logic min-sum-of-products-form numerical-answers 20

Answer key

4.26.3 Min Sum of Products Form: GATE CSE 2001 | Question: 10



- a. Is the 3-variable function $f = \sum(0, 1, 2, 4)$ its self-dual? Justify your answer.
 b. Give a minimal product-of-sum form of the b output of the following excess-3 to BCD converter.



gatecse-2001 digital-logic normal descriptive min-sum-of-products-form

Answer key 2

4.26.4 Min Sum of Products Form: GATE CSE 2005 | Question: 18

The switching expression corresponding to $f(A, B, C, D) = \Sigma(1, 4, 5, 9, 11, 12)$ is: 3

- A. $BC'D' + A'C'D + AB'D$ B. $ABC' + ACD + B'C'D$ 4
C. $ACD' + A'BC' + AC'D'$ D. $A'BD + ACD' + BCD'$

gatecse-2005 digital-logic normal min-sum-of-products-form 5

Answer key

4.26.5 Min Sum of Products Form: GATE CSE 2007 | Question: 9

Consider the following Boolean function of four variables: 6

$$f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$$
 7

The function is 8

- A. independent of one variables. B. independent of two variables 9
C. independent of three variables. D. dependent on all variables

gatecse-2007 digital-logic normal min-sum-of-products-form k-map 10

Answer key

4.26.6 Min Sum of Products Form: GATE CSE 2011 | Question: 14

The simplified SOP (Sum of Product) from the Boolean expression 11

$$(P + \bar{Q} + R). (P + \bar{Q} + R). (P + Q + R)$$
 12

is 13

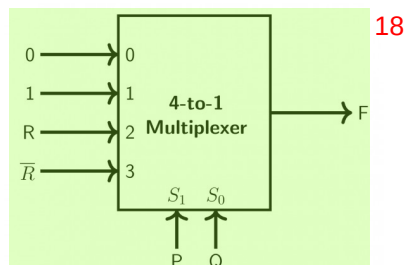
- A. $(\bar{P}.Q + \bar{R})$ B. $(P + \bar{Q}. \bar{R})$ C. $(\bar{P}.Q + R)$ D. $(P.Q + R)$ 14

gatecse-2011 digital-logic normal min-sum-of-products-form 15

Answer key

4.26.7 Min Sum of Products Form: GATE CSE 2014 | Set 1 | Question: 45

Consider the 4-to-1 multiplexer with two select lines S_1 and S_0 given below 17



The minimal sum-of-products form of the Boolean expression for the output F of the multiplexer is 19

- A. $\bar{P}Q + Q\bar{R} + P\bar{Q}R$ B. $\bar{P}Q + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}R$ 20
C. $\bar{P}QR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$ D. $PQ\bar{R}$

gatecse-2014-set1 digital-logic normal multiplexer min-sum-of-products-form 21

Answer key

4.26.8 Min Sum of Products Form: GATE CSE 2014 | Set 1 | Question: 7



Consider the following Boolean expression for F : 1

$$F(P, Q, R, S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$$

The minimal sum-of-products form of F is

A. $PQ + QR + QS$

B. $P + Q + R + S$ 2

C. $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$

D. $\bar{P}R + \bar{R}\bar{P}S + P$

gatecse-2014-set1 digital-logic normal min-sum-of-products-form

Answer key

4.26.9 Min Sum of Products Form: GATE CSE 2014 | Set 3 | Question: 7



Consider the following minterm expression for F : 3

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$
 4

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimal sum-of-products form for F is 5

A. $Q\bar{S} + \bar{Q}S$

B. $\bar{Q}\bar{S} + QS$ 6

C. $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$

D. $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

gatecse-2014-set3 digital-logic min-sum-of-products-form normal 7

Answer key

4.26.10 Min Sum of Products Form: GATE CSE 2018 | Question: 49



Consider the minterm list form of a Boolean function F given below. 8

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$
 9

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is ____ 10

gatecse-2018 digital-logic min-sum-of-products-form k-map numerical-answers two-marks 11

Answer key

4.26.11 Min Sum of Products Form: GATE CSE 2021 | Set 2 | Question: 52



Consider a Boolean function $f(w, x, y, z)$ such that 12

$$f(w, 0, 0, z) = 1$$
 13

$$f(1, x, 1, z) = x + z$$

$$f(w, 1, y, z) = wz + y$$

The number of literals in the minimal sum-of-products expression of f is ____ 14

gatecse-2021-set2 digital-logic boolean-algebra min-sum-of-products-form numerical-answers two-marks 15

Answer key

4.26.12 Min Sum of Products Form: GATE CSE 2024 | Set 1 | Question: 37



Consider a Boolean expression given by $F(X, Y, Z) = \sum(3, 5, 6, 7)$. 17

Which of the following statements is/are CORRECT?

A. $F(X, Y, Z) = \Pi(0, 1, 2, 4)$

B. $F(X, Y, Z) = X\bar{Y} + YZ + XZ$ 18

C. $F(X, Y, Z)$ is independent of input Y

D. $F(X, Y, Z)$ is independent of input X

gatecse-2024-set1 multiple-selects digital-logic min-sum-of-products-form two-marks 19

4.26.13 Min Sum of Products Form: GATE CSE 2025 | Set 1 | Question: 32



Consider the following four variable Boolean function in sum-of-product form ¹

$$F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12) \quad 2$$

where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F ? ³

- A. $\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0 + b_1\bar{b}_2b_3$ B. $\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0$ ⁴
 C. $\bar{b}_2\bar{b}_0 + b_1b_2b_3$ D. $\bar{b}_0\bar{b}_2 + \bar{b}_3$

gatecse2025-set1 digital-logic min-sum-of-products-form easy two-marks ⁵

4.26.14 Min Sum of Products Form: GATE CSE 2026 | Set 1 | Question: 38



Consider a Boolean function F with the following minterm expression: ⁶

$$F(P, Q, R, S) = \sum m(1, 2, 3, 4, 5, 7, 10, 12, 13, 14) \quad 7$$

Which of the following options is/are the minimal sum-of-products expression(s) of F ? ⁸

- A. $\bar{P}S + QR + \bar{P}\bar{Q}R + \bar{Q}R\bar{S}$ ⁹
 B. $\bar{P}S + QR + \bar{P}\bar{Q}R + PR\bar{S}$
 C. $\bar{P}S + QR + PQ\bar{S} + PR\bar{S}$
 D. $\bar{P}S + QR + PQ\bar{S} + \bar{Q}R\bar{S}$

gatecse-2026-set1 two-marks digital-logic min-sum-of-products-form k-map multiple-selects ¹⁰

4.26.15 Min Sum of Products Form: GATE CSE 2026 | Set 2 | Question: 30



Consider the following 4-variable Boolean function ¹¹

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 8, 9, 10, 11) \quad 12$$

Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function? ¹³

Note: $+$ is OR operation, $\cdot$ is AND operation, $'$ is NOT operation ¹⁴

- A. $A' + B' + C' + D'$ B. B' ¹⁵
 C. $A' \cdot B' + A \cdot B$ D. A'

gatecse-2026-set2 digital-logic min-sum-of-products-form k-map two-marks ¹⁶

4.26.16 Min Sum of Products Form: GATE IT 2008 | Question: 8



Consider the following Boolean function of four variables ¹⁸

$$f(A, B, C, D) = \sum(2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$$

The function is

- A. independent of one variable B. independent of two variables ¹⁹
 C. independent of three variable D. dependent on all the variables

gateit-2008 digital-logic normal min-sum-of-products-form ²⁰

Practice Tests: Test 1 (15Q) Test 2 (13Q) Weekly Quiz 4 (15Q) 1

4.27.1 Multiplexer: GATE CSE 1990 | Question: 5-b



Show with the help of a block diagram how the Boolean function : 2

$$f = AB + BC + CA \quad 3$$

can be realised using only a 4 : 1 multiplexer. 4

gate1990 descriptive digital-logic combinational-circuit multiplexer 5

Answer key 6

4.27.2 Multiplexer: GATE CSE 1998 | Question: 1.14



A multiplexer with a 4 – *bit* data select input is a 7

- A. 4 : 1 multiplexer
C. 16 : 1 multiplexer
B. 2 : 1 multiplexer 8
D. 8 : 1 multiplexer

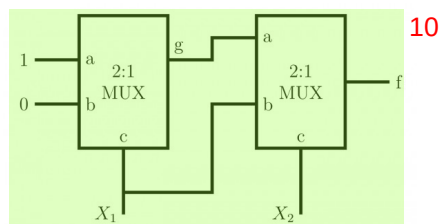
gate1998 digital-logic multiplexer easy

Answer key

4.27.3 Multiplexer: GATE CSE 2001 | Question: 2.11



Consider the circuit shown below. The output of a 2 : 1 MUX is given by the function $(ac' + bc)$. 9



Which of the following is true? 11

- A. $f = X_1' + X_2$
C. $f = X_1 X_2 + X_1' X_2'$
B. $f = X_1' X_2 + X_1 X_2'$ 12
D. $f = X_1 + X_2'$

gatecse-2001 digital-logic normal multiplexer 13

Answer key

4.27.4 Multiplexer: GATE CSE 2004 | Question: 60



Consider a multiplexer with X and Y as data inputs and Z the as the control input. $Z = 0$ selects input X , and $Z = 1$ selects input Y . What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware? 14

- A. R to X, 1 to Y, T to Z
C. T to X, R to Y, 0 to Z
B. T to X, R to Y, T to Z 15
D. R to X, 0 to Y, T to Z

gatecse-2004 digital-logic normal multiplexer 16

Answer key

4.27.5 Multiplexer: GATE CSE 2007 | Question: 34



Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed? 17

- A. 2^n line to 1 line
C. 2^{n-1} line to 1 line
B. 2^{n+1} line to 1 line 19
D. 2^{n-2} line to 1 line

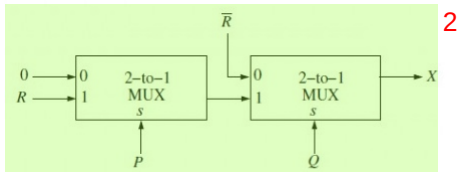
gatecse-2007 digital-logic normal multiplexer 20

Answer key

4.27.6 Multiplexer: GATE CSE 2016 | Set 1 | Question: 30



Consider the two cascade 2 to 1 multiplexers as shown in the figure . 1



The minimal sum of products form of the output X is 3

A. $\bar{P}\bar{Q} + PQR$

B. $\bar{P}Q + QR$ 4

C. $PQ + \bar{P}\bar{Q}R$

D. $\bar{Q}\bar{R} + PQR$

gatecse-2016-set1 digital-logic multiplexer normal 5

Answer key

4.27.7 Multiplexer: GATE CSE 2020 | Question: 19



A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The number of select lines needed for the multiplexer is _____. 5

gatecse-2020 numerical-answers digital-logic multiplexer one-mark 7

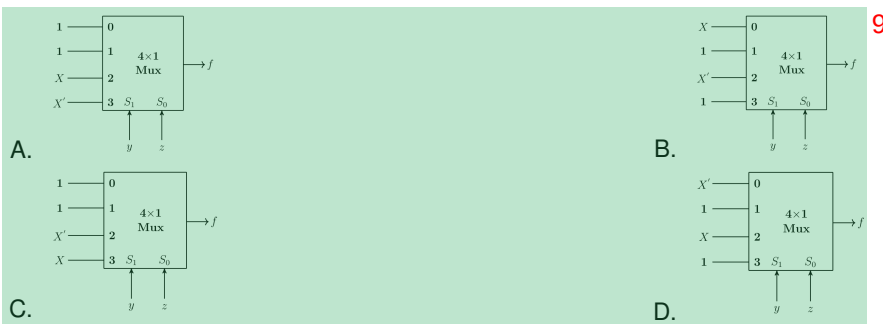
Answer key

4.27.8 Multiplexer: GATE CSE 2021 | Set 2 | Question: 5



Which one of the following circuits implements the Boolean function given below? 8

$f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm.



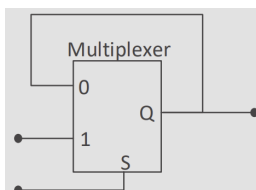
gatecse-2021-set2 digital-logic combinational-circuit multiplexer one-mark 10

Answer key

4.27.9 Multiplexer: GATE CSE 2023 | Question: 11



The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.



Match the functional equivalence of this circuit to one of the following options.

A. D Flip-flop

B. D Latch

C. Half-adder

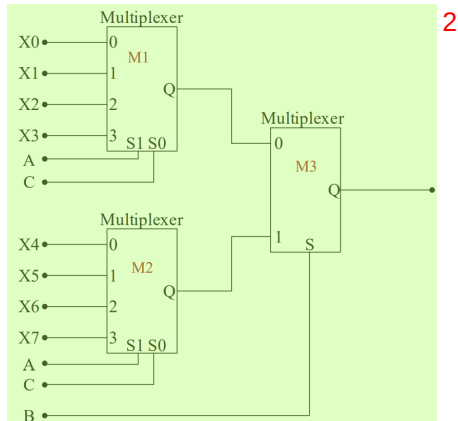
D. Demultiplexer

gatecse-2023 digital-logic combinational-circuit multiplexer one-mark

4.27.10 Multiplexer: GATE CSE 2023 | Question: 34



A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0-X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $\bar{A} + \bar{A} \cdot \bar{C} + A \cdot \bar{B} \cdot C$?

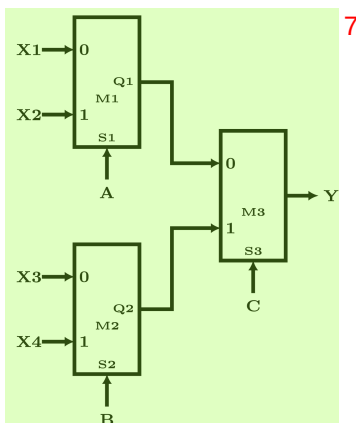
- A. (1,1,0,0,1,1,1,0) B. (1,1,0,0,1,1,0,1)
C. (1,1,0,1,1,1,0,0) D. (0,0,1,1,0,1,1,1)

gatescse-2023 digital-logic combinational-circuit multiplexer two-marks

4.27.11 Multiplexer: GATE CSE 2024 | Set 1 | Question: 54



Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.



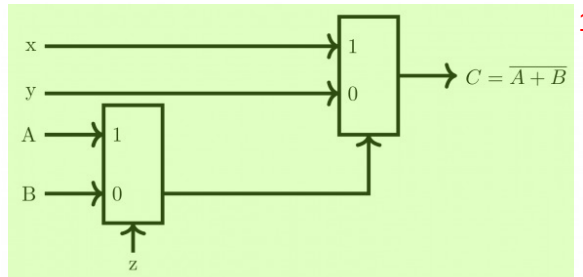
For an instance of inputs $X1 = 1, X2 = 1, X3 = 0$, and $X4 = 0$, the number of combinations of A, B, C that give the output $Y = 1$ is _____.

gatescse-2024-set1 numerical-answers digital-logic multiplexer two-marks

4.27.12 Multiplexer: GATE IT 2005 | Question: 48



The circuit shown below implements a 2-input NOR gate using two 2 – 4 MUX (control signal 1 selects the upper input). What are the values of signals x, y and z?



1

- A. 1,0,B B. 1,0,A C. 0,1,B D. 0,1,A

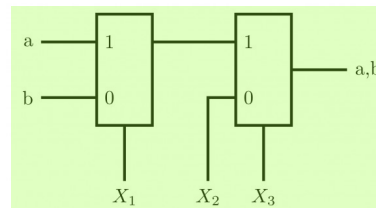
2

gateit-2005 digital-logic normal multiplexer 3

Answer key

4.27.13 Multiplexer: GATE IT 2007 | Question: 8

The following circuit implements a two-input AND gate using two 2 – 1 multiplexers.



5

What are the values of X_1, X_2, X_3 ?

- A. $X_1 = b, X_2 = 0, X_3 = a$ B. $X_1 = b, X_2 = 1, X_3 = b$
 C. $X_1 = a, X_2 = b, X_3 = 1$ D. $X_1 = a, X_2 = 0, X_3 = b$

7

gateit-2007 digital-logic normal multiplexer 8

Answer key

4.27.14 Multiplexer: GATE1992-04-b

A priority encoder accepts three input signals (A, B and C) and produces a two-bit output (X_1, X_0) corresponding to the highest priority active input signal. Assume A has the highest priority followed by B and C has the lowest priority. If none of the inputs are active the output should be 00, design the priority encoder using 4 : 1 multiplexers as the main components.

10

gate1992 digital-logic combinational-circuit multiplexer descriptive 11

Answer key

4.28 Number Representation (57)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (9Q) 12

4.28.1 Number Representation: GATE CSE 1990 | Question: 1-viii

The condition for overflow in the addition of two 2's complement numbers in terms of the carry generated by the two most significant bits is _____.

gate1990 digital-logic number-representation fill-in-the-blanks

Answer key

4.28.2 Number Representation: GATE CSE 1991 | Question: 01-iii

Consider the number given by the decimal expression:

$$16^3 * 9 + 16^2 * 7 + 16 * 5 + 3$$

The number of 1's in the unsigned binary representation of the number is _____

Answer key

4.28.3 Number Representation: GATE CSE 1991 | Question: 01-v

When two 4-bit numbers $A = a_3a_2a_1a_0$ and $B = b_3b_2b_1b_0$ are multiplied, the bit c_1 of the product C is given by _____

gate1991 digital-logic normal number-representation fill-in-the-blanks 2

Answer key

4.28.4 Number Representation: GATE CSE 1992 | Question: 4-a

Consider addition in two's complement arithmetic. A carry from the most significant bit does not always correspond to an overflow. Explain what is the condition for overflow in two's complement arithmetic.

gate1992 digital-logic normal number-representation descriptive 4

Answer key

4.28.5 Number Representation: GATE CSE 1993 | Question: 6.5

Convert the following numbers in the given bases into their equivalents in the desired bases: 5

- A. $(110.101)_2 = (x)_{10}$
 B. $(1118)_{10} = (y)_H$

gate1993 digital-logic number-representation normal descriptive 7

Answer key

4.28.6 Number Representation: GATE CSE 1994 | Question: 2.7

Consider n -bit (including sign bit) 2 's complement representation of integer numbers. The range of integer values, N , that can be represented is _____ $\leq N \leq$ _____.

gate1994 digital-logic number-representation easy fill-in-the-blanks 9

Answer key

4.28.7 Number Representation: GATE CSE 1995 | Question: 18

The following is an incomplete Pascal function to convert a given decimal integer (in the range -8 to $+7$) into a binary integer in 2 's complement representation. Determine the expressions A , B , C that complete program.

```
function TWOSCOMP (N:integer):integer;
var
  REM, EXPONENT:integer;
  BINARY :integer;
begin
  if (N>=-8) and (N<=+7) then
  begin
    if N<0 then
      N:=A;
    BINARY:=0;
    EXPONENT:=1;
    while N<>0 do
    begin
      REM:=N mod 2;
      BINARY:=BINARY + B*EXPONENT;
      EXPONENT:=EXPONENT*10;
      N:=C
    end
    TWOSCOMP:=BINARY
  end
end;
```

gate1995 digital-logic number-representation normal descriptive

Answer key

4.28.8 Number Representation: GATE CSE 1995 | Question: 2.12, ISRO2015-9



The number of 1's in the binary representation of $(3 * 4096 + 15 * 256 + 5 * 16 + 3)$ are: ¹

- A. 8 B. 9 C. 10 D. 12 ²

gate1995 digital-logic number-representation normal isro2015 ³

Answer key

4.28.9 Number Representation: GATE CSE 1996 | Question: 1.25



Consider the following floating-point number representation. ⁴

31	24	23	0
Exponent		Mantissa	

The exponent is in 2 's complement representation and the mantissa is in the sign-magnitude representation. The range of the magnitude of the normalized numbers in this representation is ⁶

- A. 0 to 1 B. 0.5 to 1 C. 2^{-23} to 0.5 D. 0.5 to $(1 - 2^{-23})$ ⁷

gate1996 digital-logic number-representation normal ⁸

Answer key

4.28.10 Number Representation: GATE CSE 1997 | Question: 5.4



Given $\sqrt{(224)_r} = (13)_r$. ¹⁰

The value of the radix r is:

- A. 10 B. 8 C. 5 D. 6 ¹¹

gate1997 digital-logic number-representation normal ¹²

Answer key

4.28.11 Number Representation: GATE CSE 1998 | Question: 1.17



The octal representation of an integer is $(342)_8$. If this were to be treated as an eight-bit integer in an 8085-based computer, its decimal equivalent is ¹³

- A. 226 B. -98 C. 76 D. -30 ¹⁵

gate1998 digital-logic number-representation normal ¹⁶

Answer key

4.28.12 Number Representation: GATE CSE 1998 | Question: 2.20



Suppose the domain set of an attribute consists of signed four digit numbers. What is the percentage of reduction in storage space of this attribute if it is stored as an integer rather than in character form? ¹⁷

- A. 80\% B. 20\% C. 60\% D. 40\% ¹⁹

gate1998 digital-logic number-representation normal ²⁰

Answer key

4.28.13 Number Representation: GATE CSE 1999 | Question: 2.17



Zero has two representations in ²²

- A. Sign-magnitude B. 2 's complement ²³
C. 1 's complement D. None of the above

gate1999 digital-logic number-representation easy multiple-selects ²⁴

Answer key